Quantitative Forest Genetics and Genomics: from linear mixed models to AI-powered prediction

Twenty years of breeding value prediction through increasingly biologically informed models

Eduardo Pablo Cappa

E-mail: *cappa.eduardo@inta.gob.ar*

INTA – CONICET – UBC

Centro de Investigación en Recursos Naturales

Instituto de Recursos Biológicos

Bosques Cultivados

THE UNIVERSITY OF BRITISH COLUMBIA

CONICET

Piracicaba, San Pablo, Brasil
October 6 - 8th, 2026

Instituto Nacional de Tecnología Agropecuaria

Ministerio de Economía República Argentina

Secretaría de Agricultura, Ganadería y Pesca

General goal

To present the evolution of predictive models in forest tree breeding over the last two decades, highlighting how the progressive integration of biological information -from pedigree and genomics to enviromics, multi-omics and AI- has transformed breeding value prediction.

Presentation Roadmap



PART I

Foundations of Prediction

From classical models to genomics

1. The Big Picture
2. Spatial & Competition Models
3. From Pedigree to Genomics

PART II

Making Genomic Selection Operational

From genomic prediction to real breeding programs

4. Operational Genomic Selection
5. Training Populations & Prediction Across Generations
6. Identifying the Right Trees

PART III

Increasing Biological Complexity

From correlations to biological mechanisms

7. Multi-trait Prediction
8. From Correlation to Causation
9. Machine Learning for Genomic Prediction

PART IV

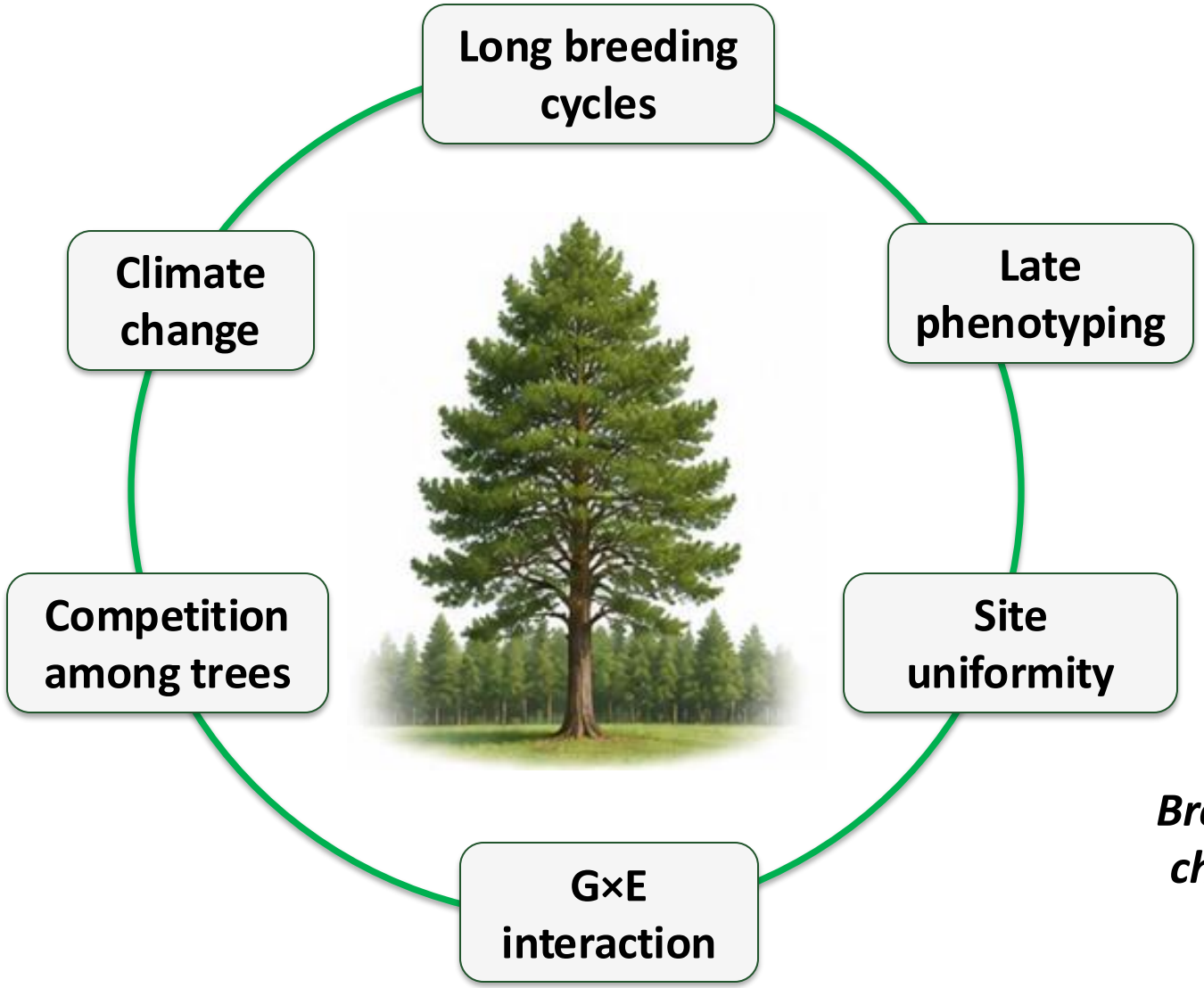
Integrating Environment and Biology

Environment, multi-omics and future systems

10. Prediction Across Environments
11. Environmental & Biological Covariates
12. **Looking ahead in forest tree breeding**

Why is forest tree breeding so challenging?

Forest tree breeding is one of the most difficult prediction problems in quantitative genetics



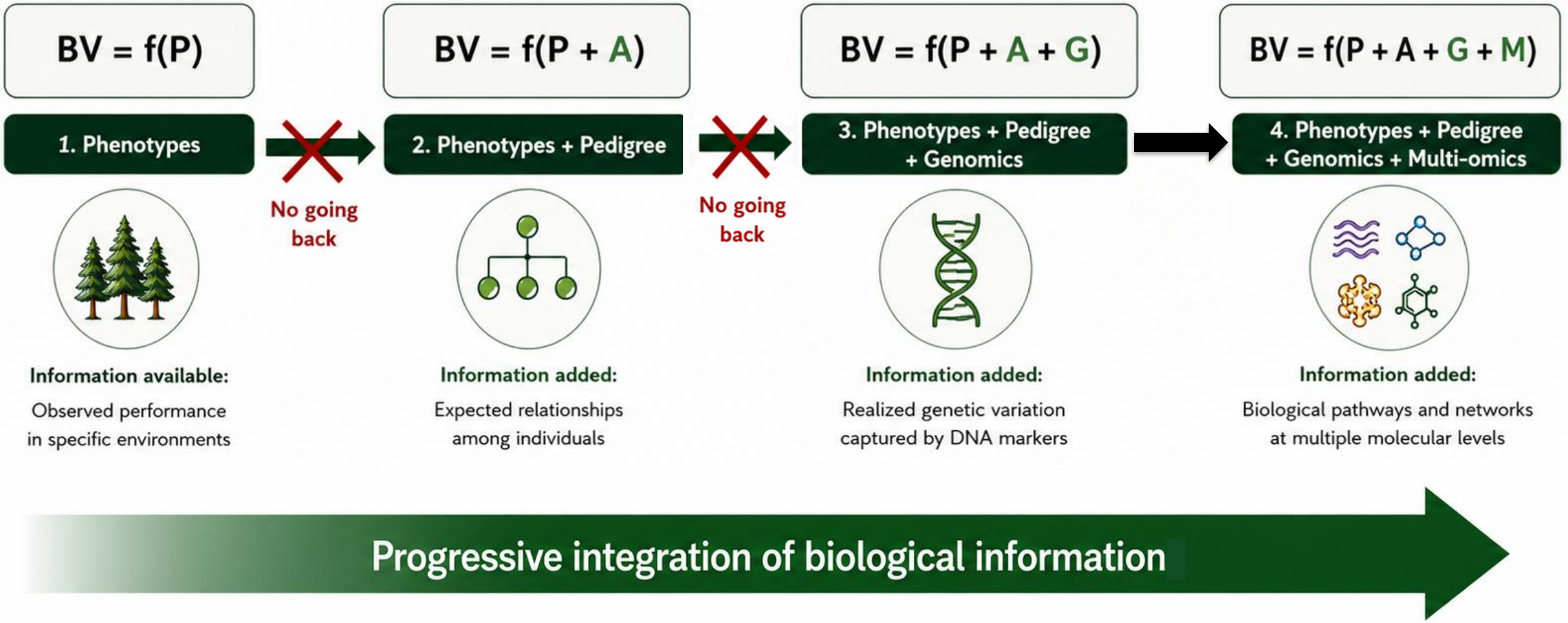
Key challenges

- Trial size: experimental units - area.
- Multi-site field trials
- Multi-trait evaluation
- Repeated measurements over time
- Multi-generations and populations

Breeding value prediction remains a major challenge in forestry genetics, driving the evolution of predictive models.

Progressive integration of biological information

Over the last decades, forest tree breeders have progressively incorporated new sources of biological information into their prediction models.



What limitation was each method trying to solve?

Looking back, every methodological advance has emerged as a response to a biological limitation.

Limitation	Methodological advance
Environmental heterogeneity	Spatial models
Tree interactions	Competition models
Pedigree-based expected relationships	Genomics
Independent traits and environments	Multi-trait & multi-environment G×E models
Correlation only	Causal models
Incomplete biological interpretation	Multi-omics
Linear assumptions	Machine Learning

Central hypothesis

Better predictions emerge as models progressively capture biological reality

A broader trend across breeding and biology

Trends in Plant Science
CellPress OPEN ACCESS

Opinion

Better breeding leveraging more biology

Owen M. Powell [1,2,3,*], Lee Hickey [1,2,3], Shunichiro Tomura [1,2,3], and Mark Cooper [1,2,3]

Climate-driven variability is reducing our ability to accurately predict crop performance across environments, limiting genetic gain in breeding programs. Sustained progress requires predictive frameworks that capture plant–environment interactions across diverse genetics and management conditions. Integrating mechanistic insights from plant science into predictive models offers a path to improve the accuracy, precision, and interpretability of breeding grounds under cracy, precision, and interpretability of breeding hierachical genome-pimige environments and outline how they can be lereraged within breeding programs to evalaate how biological knowledge informs predictions across target environments and supports long-term genetic gain.

Highlights

Increasing climate-driven variability of cropping systems is testing the limits of current statistical models to predict crop performance across environments.

To improve long-term genetic gain of breeding programs, prediction models [illegible] interactions across diverse genetic backgrounds, environmental conditions and management practices.

CellPress Partner Journal
Molecular Plant
Opinion

Biology-driven genebank pangenetics meet AI-powered predictions to transform crop breeding

The modern era of crop improvement is characterized by two distinct paths: (i) biology-driven molecular genetics focused on dissecting the mechanisms underlying complex traits by a reductionist approach and (ii) data-driven quantitative genetics, which prioritizes accurate prediction of final trait performance, such as yield, without necessarily clarifying the underlying biological mechanisms. These approaches, one [illegible] statistical association, have each made significant strides, yet with limited overlap. However, their conjunction represents an emerging opportunity: integrating the two ideal crops? Following patterns or understanding the processes that shape them?

We argue that new opportunities for crop improvement lie in striking a balance—combining the statistical power and scalability of quantitative genetics with causal understanding derived from molecular genetics (Figure 1). But how? Here, we discuss progress in molecular and quantitative genetics and envision a future where these can be integrated.



Cell
Leading Edge
CellPress

Perspective

Why machines don't speak biology: Toward native biological language models

Yonatan Stelzer [1,*], and Amos Tanay [1,2,*]
[1]Department of Molecular Cell Biology, Weizmann Institute of Science, Rehovot, Israel
[2]Department of Computer Science and Applied Mathematics, Weizmann Institute of Science, Rehovot, Israel
*Correspondence: yonatan.stelzer@weizmann.ac.il (Y.S.), amos.tanay@weizmann.ac.il (A.T.)
https://doi.org/10.1016/j.cell.2025.07.003

SUMMARY

Recent advances in AI inspire visions of universal models of biology. Yet living systems are evolved, emergent processes whose behaviors cannot be inferred from their parts alone. We propose grounding AI in canonical biological processes, constructing data-driven world models with explicit mechanistic links across molecules, cells, and their dynamics in space and time.

Common message across disciplines

Better biological representation → Better prediction → Better breeding decisions

PART I: Spatial & Competition Models

The first limitation: heterogeneous field conditions

Cappa et al. (Sil Gen, 2015) – Tsuga heterophylla

Forest trials are **rarely** environmentally homogeneous.

Soil fertility

Moisture

Topography

Previous land use or management

These factors create spatial variation in tree growth across the trial.

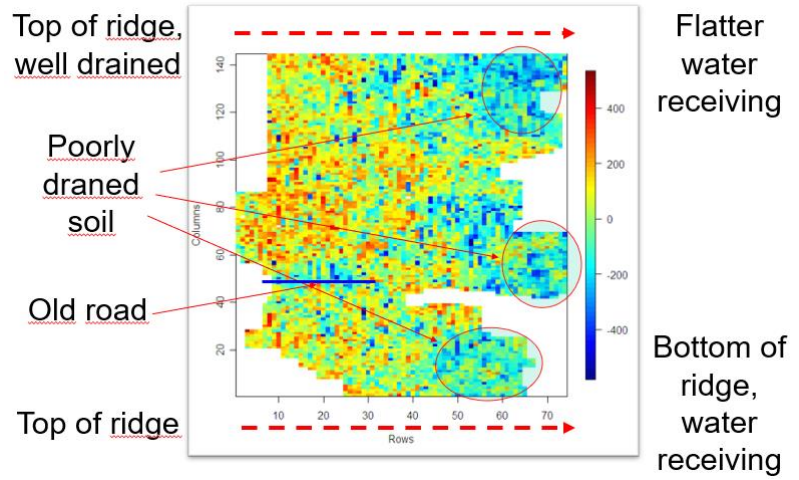
Example of a forest trial



Trial showing environmental gradients



Spatial variation (example)

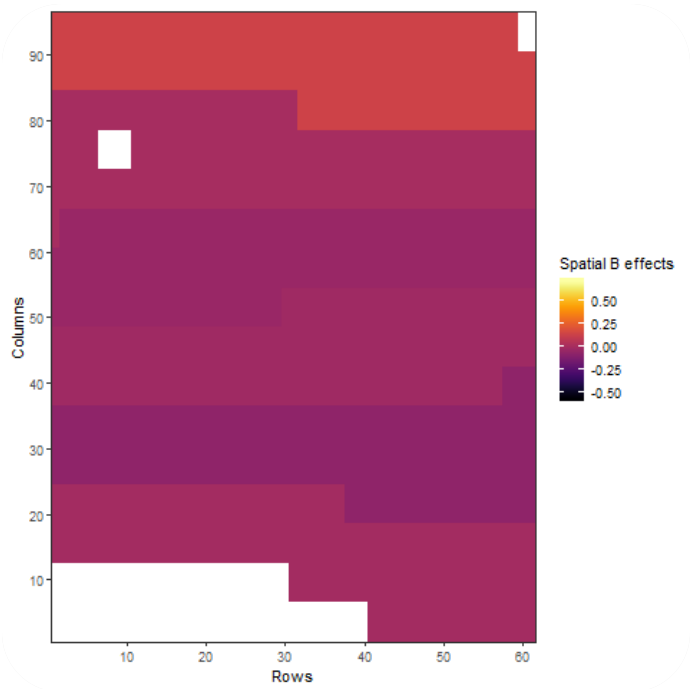


Ignoring environmental heterogeneity can **bias** the estimation of genetic parameters and the prediction of **breeding values** (i.e., BLUP of BV).

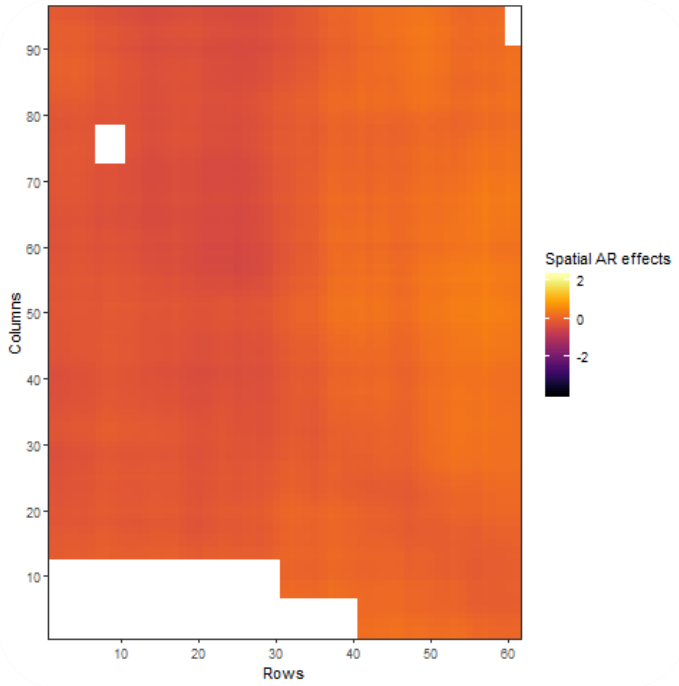
Traditional solution: experimental design

"A Priori" designs to control environmental heterogeneity: RCB or IBD

Randomized complete blocks



Spatial effects



Limitations

- Fixed block boundaries
- Cannot capture continuous spatial trends
- Environmental pattern unknown before establishment

Choosing an optimal design is often difficult or impossible.

Models must learn spatial patterns from the data, "a posteriori".

Spatial models

Spatial models

Cappa & Cantet (CJFR, 2007) – E. globulus

Several spatial approaches have been proposed to model environmental variation in field trials.

Post-blocking

Optimized incomplete blocks fitted after trial establishment.

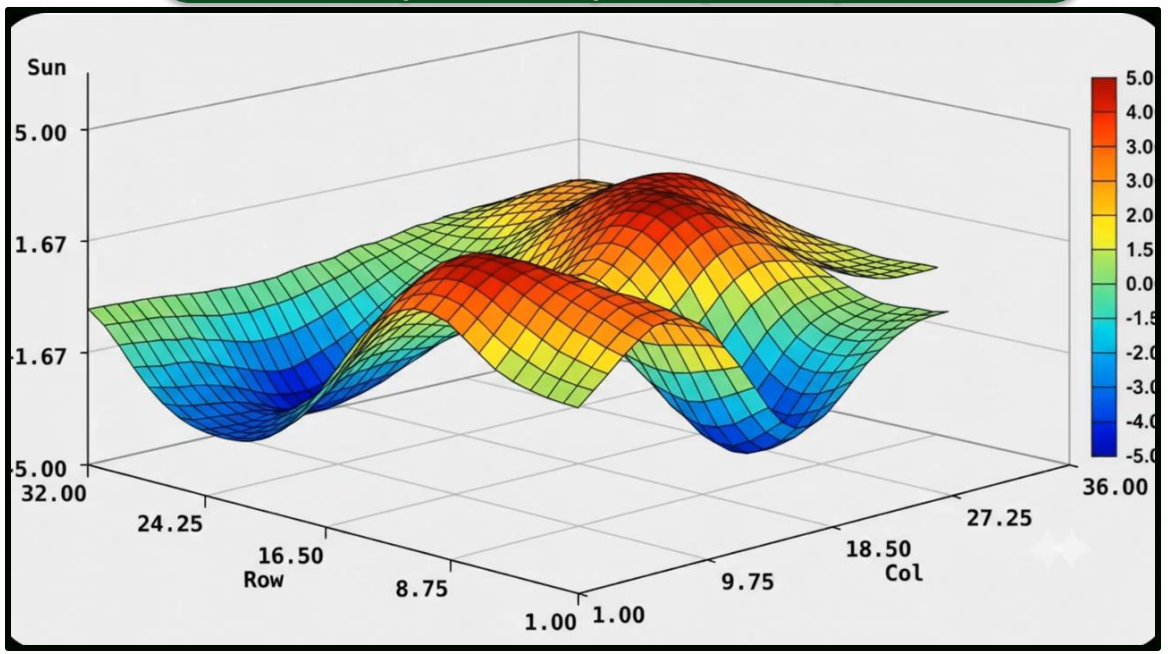
AR(1)

Autoregressive structures for rows and/or columns.

Tensor product B-splines

Two-dimensional smoothing approach based on tensor products of B-spline bases.

Example: spatial surface estimated with tensor-product B-splines (8 knots)



1 Flexible modeling of large- and small-scale variation

Tensor-product B-splines within mixed models capture complex environmental patterns in two dimensions.

2 Improved breeding value prediction

Spatial models reduce environmental noise, increasing heritability, accuracy of breeding values and expected genetic gain.

3 Easily implemented in operational tree breeding

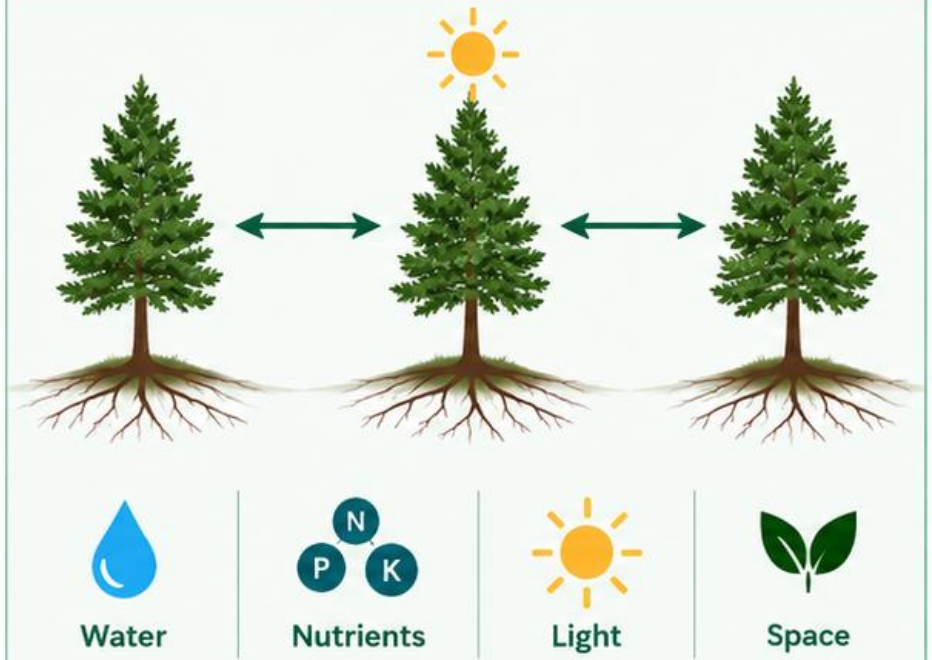
The methodology is computationally efficient and can be incorporated into routine genetic evaluation programs.

Key message

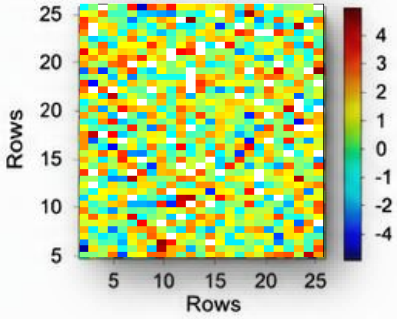
Spatial models were the first major step toward incorporating biological realism into breeding value prediction.

The second limitation: trees do not grow in isolation

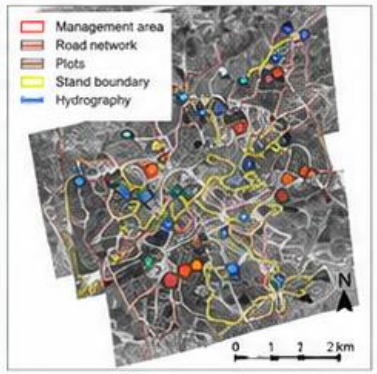
What drives competition?



Evidence from the field

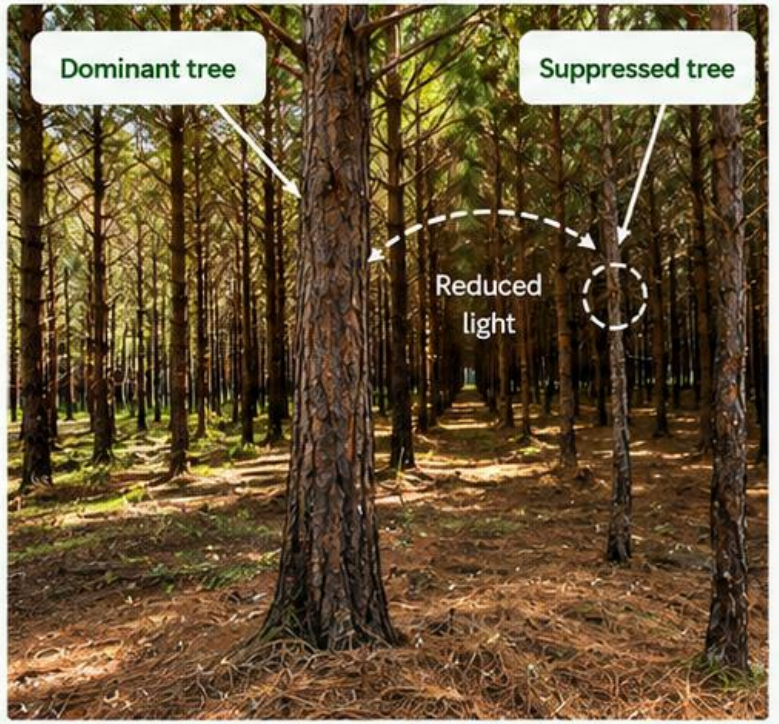


Spatial pattern of residuals DBH in the *Pinus taeda L.*



Aerial view of the root system of a hydraulically excavated clone of *Quercus pyrenaica*.

Competition in the field



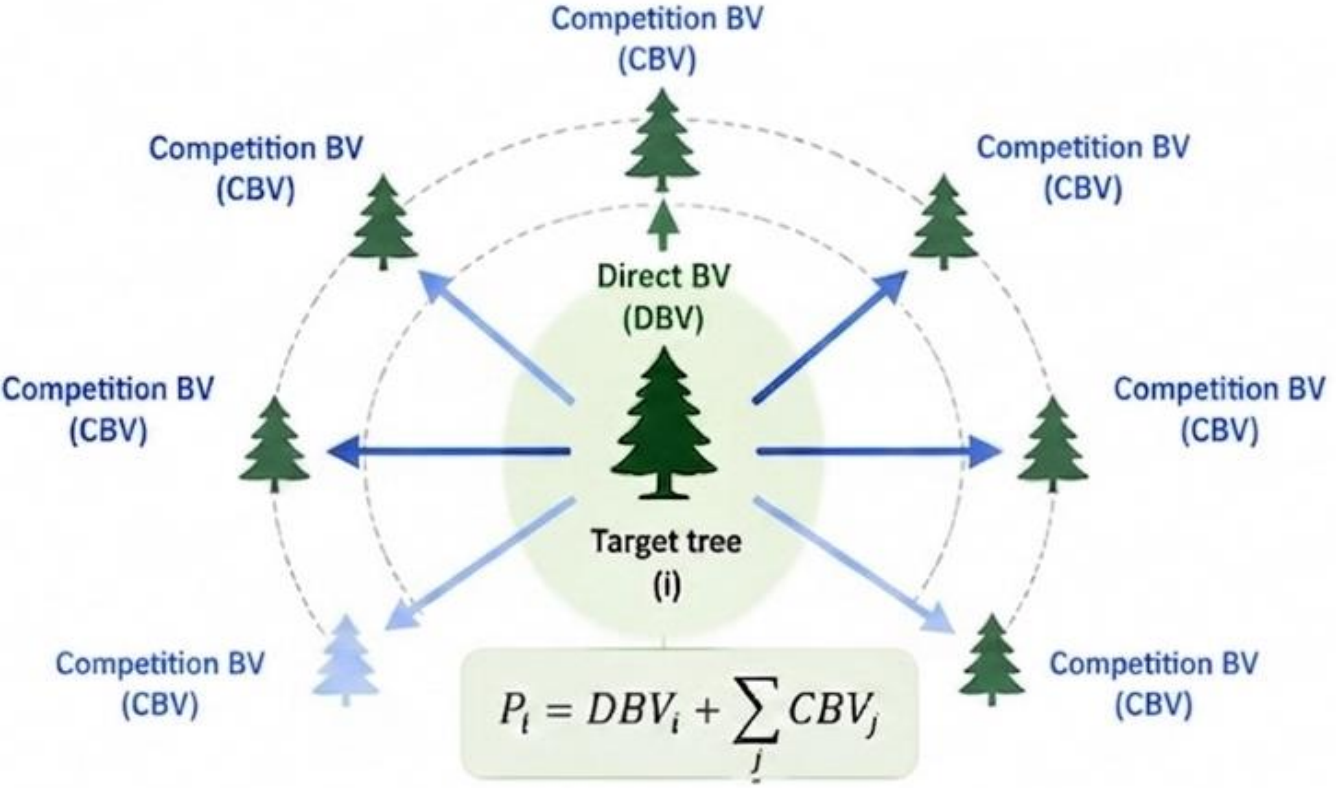
1. **Competition is caused by genetic and environmental sources**

2. **Ignoring competition effects can *bias* the estimation of genetic parameters and the prediction of breeding values (i.e., BLUP of BV).**

Competition model

Cappa & Cantet (Sil Gen, 2008) – P. taeda

To account for competition, the phenotype of an individual can be modeled as a combination of two breeding values



Direct breeding value (DBV)

- Effects expressed in the phenotype of an individual.

Competition breeding value (CBV)

- Effects expressed only in the phenotype of another individual.

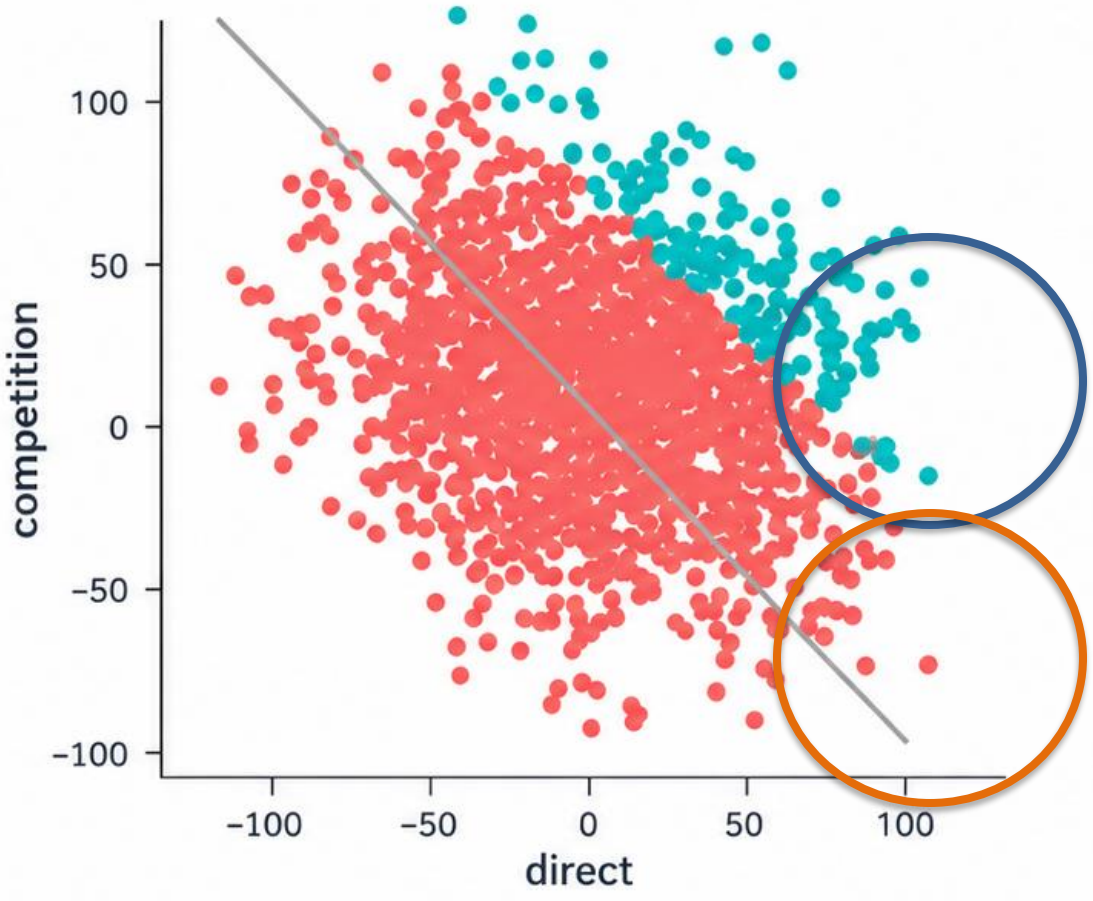
Intensity of competition (IC): modelled as an inverse function of neighbor distance and number of competing trees.



Competition model: beyond individual performance

Cappa & Cantet (Sil Gen, 2008) – P. taeda

Competition models distinguish direct and competition breeding values for selection decisions.



Traditional selection

- Selects trees with high Direct BV only.
- Tends to favour highly competitive (genotypes that suppress neighbours).

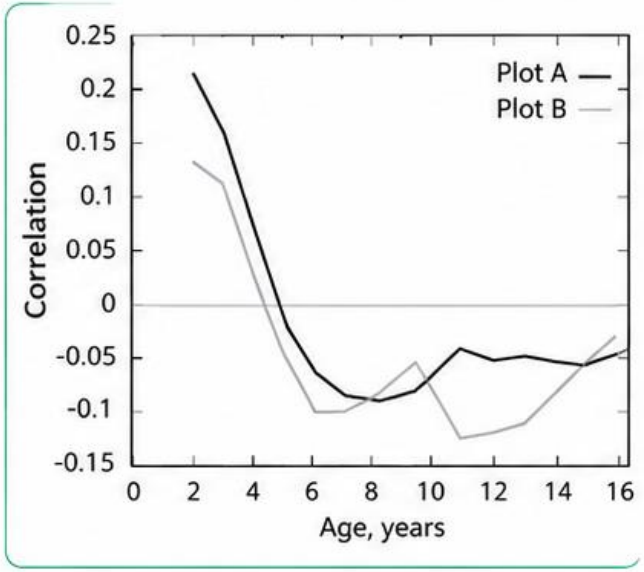
Competition-aware selection

- Selects trees using Direct BV + Competition BV.
- Identifies **productive** and **less competitive** genotypes.

Selection shifts from maximizing individual growth to maximizing **stand productivity.**

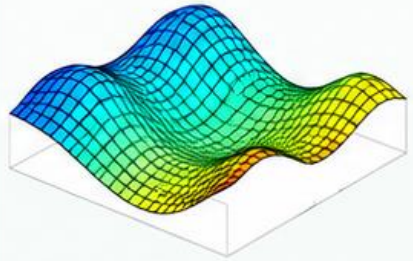
Joint modeling of spatial and competition effects

Cappa et al. (TGG, 2015) – P. taeda
Cappa et al. (TGG, 2016) – Pseudotsuga menziesii

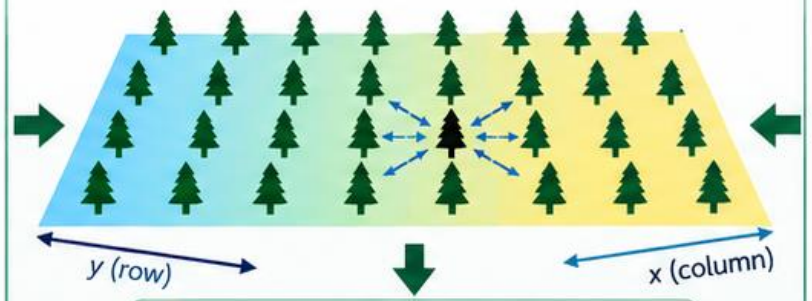


Spatial variation

Environmental heterogeneity modelled with a **two-dimensional smoothing surface.**



Forest genetic trial



CSM
Competition + Spatial Model
Accounts for both spatial variation and competition (genetic and environmental).

Competition

Direct, genetic and environmental competition effects from neighbouring trees.

- Direct, genetic effect (own tree)
- Genetic competition (neighbour genetics)
- Environmental competition (neighbour environment)

WHAT DID THE JOINT MODEL REVEAL?

1 Ignoring competition → overestimates environmental heterogeneity

Higher spatial variance when competition is ignored.

2 Ignoring environmental heterogeneity → underestimates competition effects

Underestimated direct-competition genetic correlation and environmental competition variance.

3 High BV correlations ≠ identical selection

The top 5% ranked trees changed across models (TM–CSM).

TAKE-HOME MESSAGE:

Spatial heterogeneity and competition coexist: joint modeling leads to realistic genetic evaluations and better selections.

PART I: Pedigree to Genomics

… And the genome became part of the mixed models.

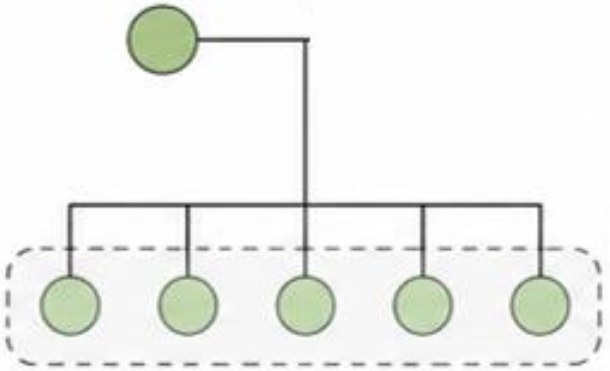
The third limitation: pedigree-based expected relationships

Cappa et al. (Mol Breed, 2016) – E. grandis

EXPECTED RELATIONSHIPS from pedigree

- Contains only expected relationships.
- Ignore historical relatedness.

Example: half-sib individuals share 25% of genes (A = 0.25)



All half-sib pairs have the same expected relationship: A = 0.25

REALIZED RELATIONSHIPS estimated with molecular markers

Marker data capture the actual proportion of genome

	1	2	3	4	5
1	–	0.18	0.22	0.31	0.19
2	0.18	–	0.27	0.16	0.21
3	0.22	0.27	–	0.23	0.24
4	0.31	0.16	0.23	–	0.20
5	0.19	0.21	0.24	0.20	–

Higher realized relationship

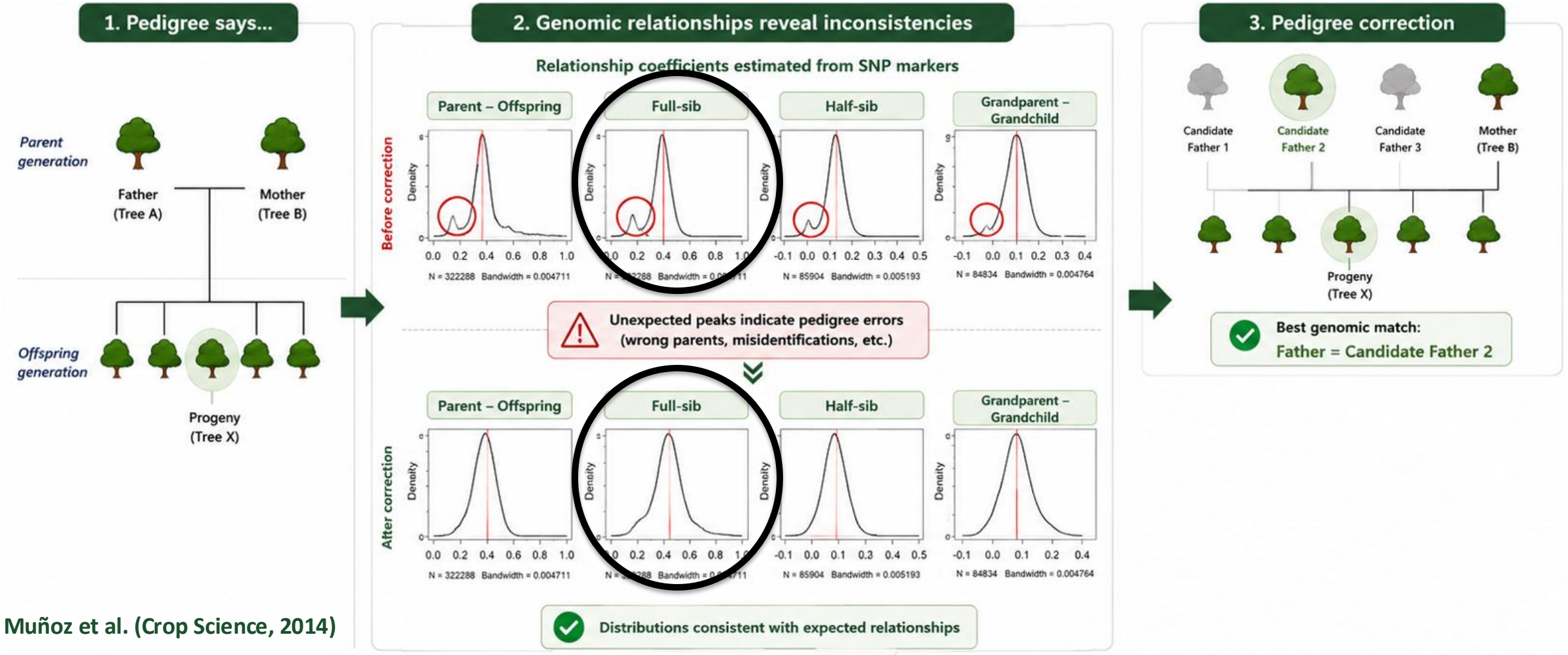
Lower realized relationship

Mendelian sampling makes every pair of relatives genetically unique.

Key message

Improve the accuracy of genetic parameter estimates and breeding value prediction.

Using genomic data to validate and correct pedigrees

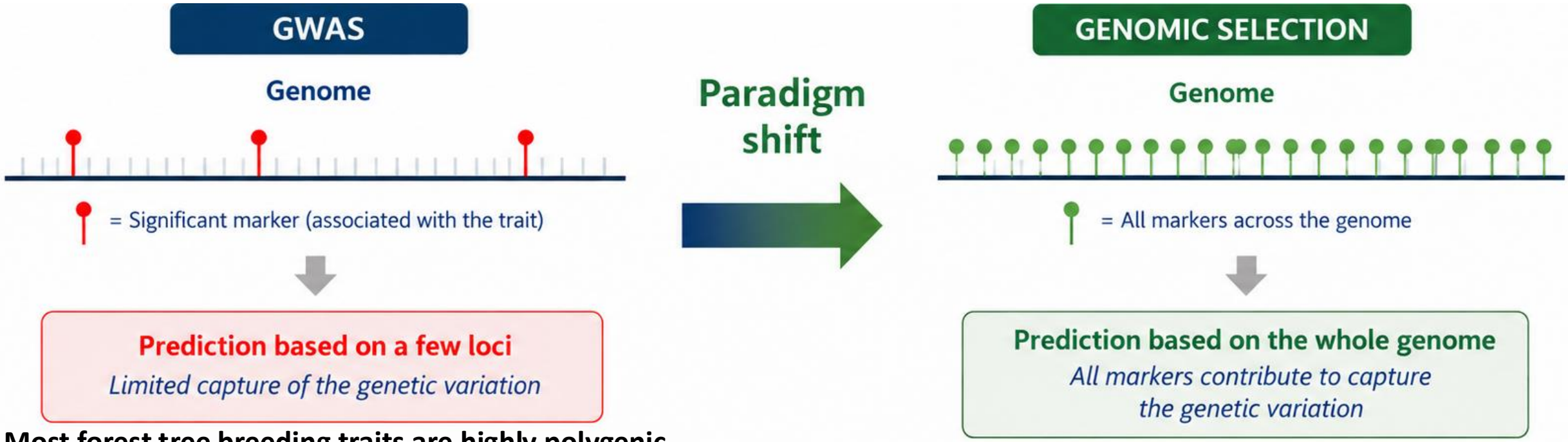


Muñoz et al. (Crop Science, 2014)

Genomic information improves pedigree accuracy by identifying and correcting relationship errors.

From Gene Discovery to Genome-wide Prediction

Cappa et al. (Plos One, 2013) – E. globulus



Most forest tree breeding traits are highly polygenic.

GWAS performs best for oligogenic traits,

Shalizi et al. (G3, 2021) – P. taeda – fusiform rust

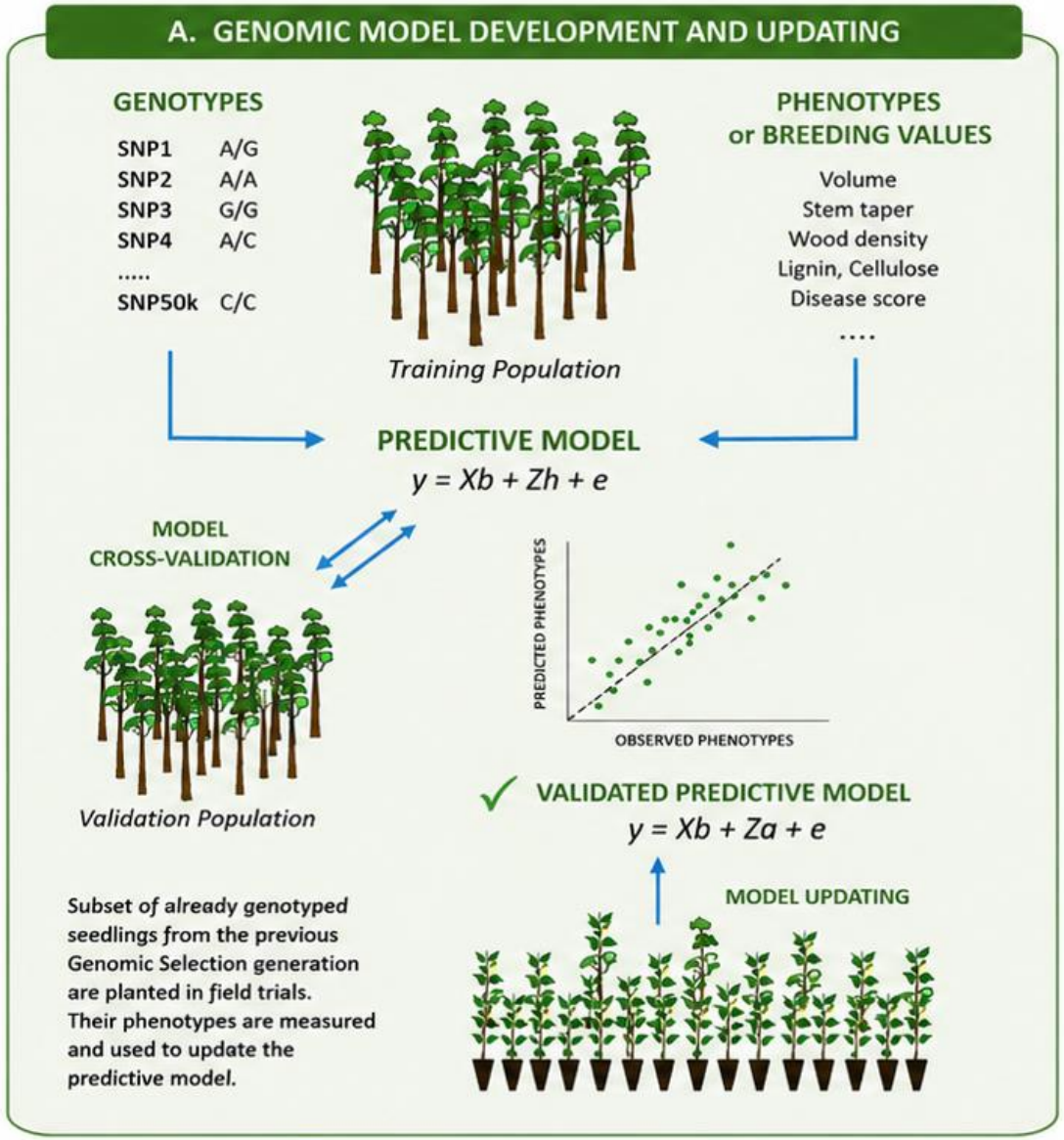
The question changed from "Which genes matter?" to "How well can we predict?"

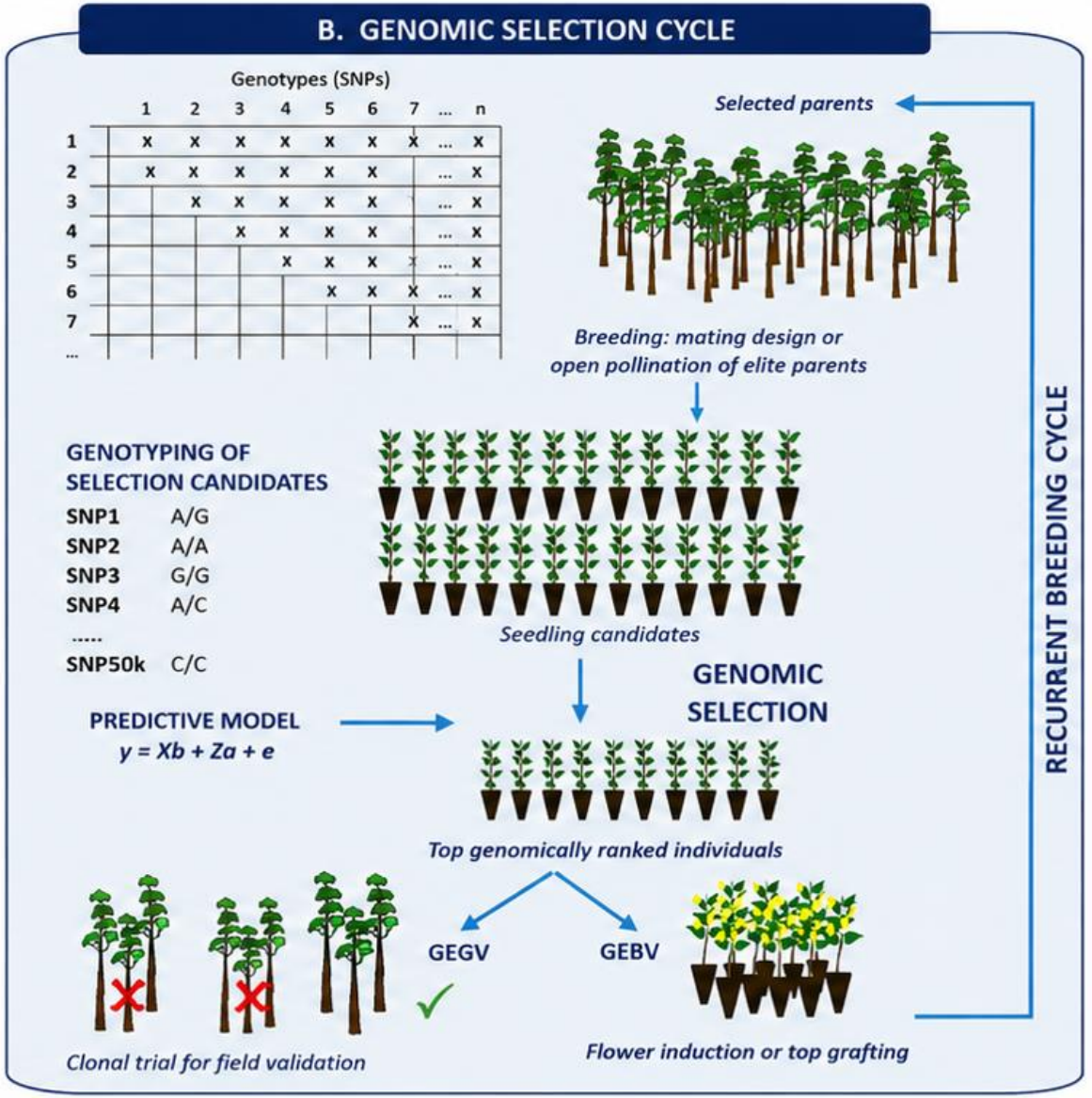
GWAS
Gene discovery

Genomic Selection
Genome-wide prediction

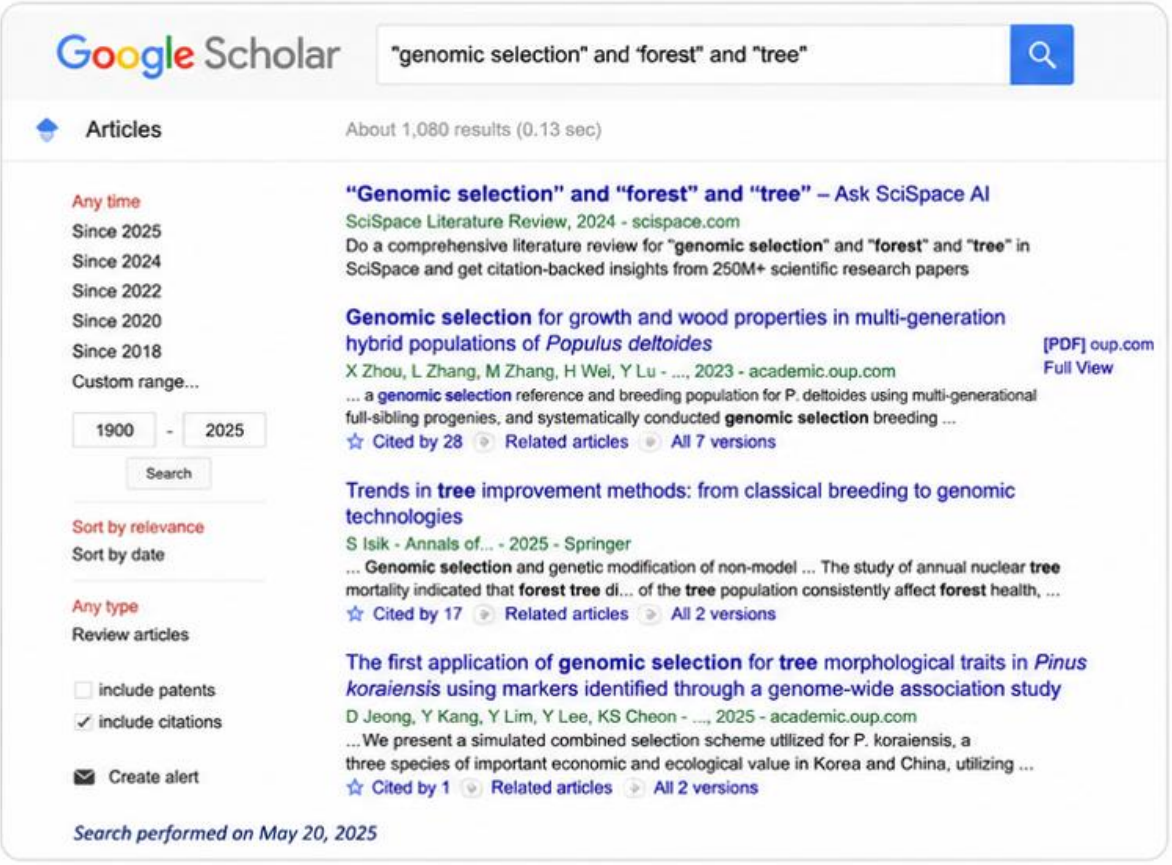
How genomic selection works

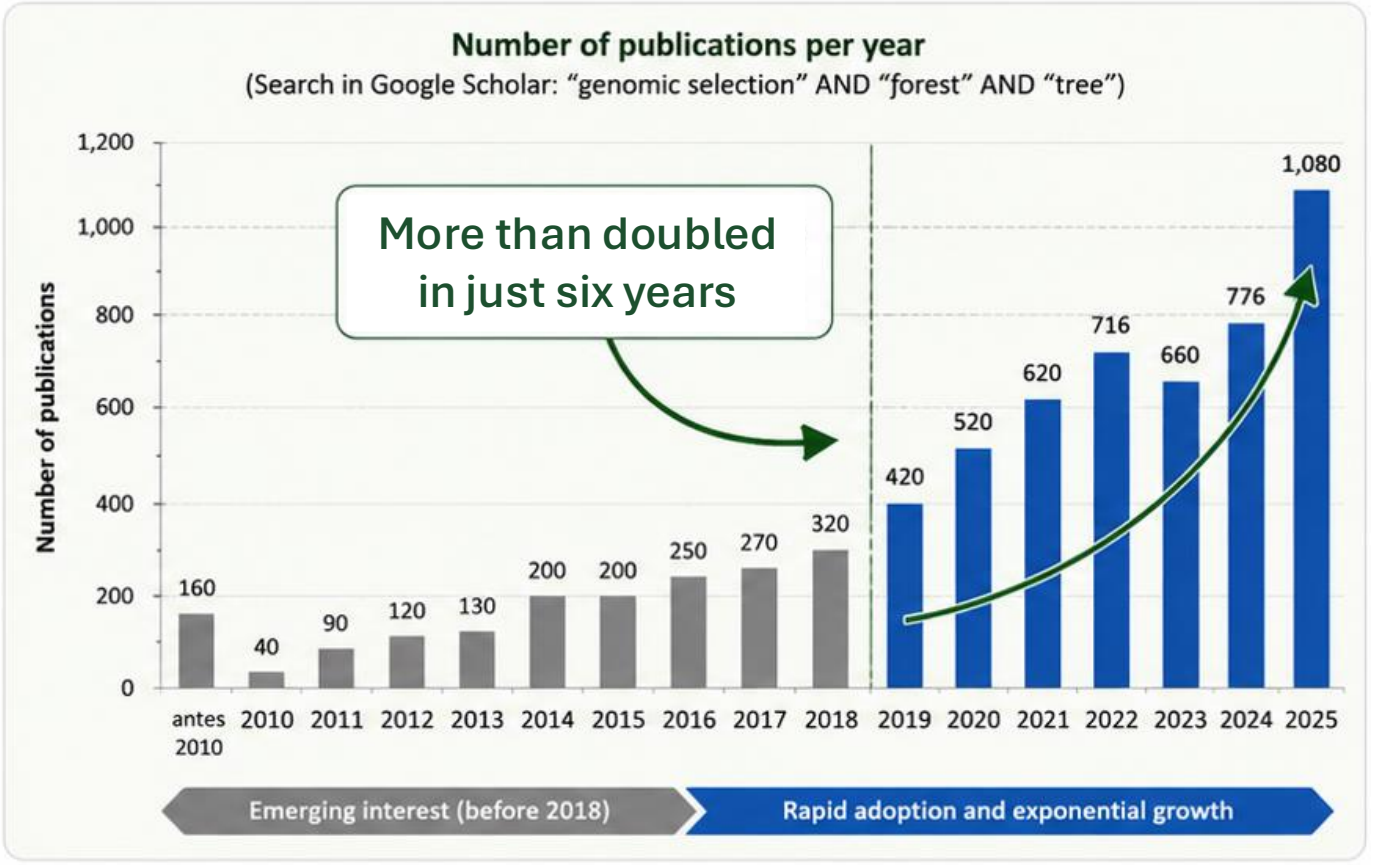
Grattapaglia et al. (Frontiers in Plant Sci, 2018)





The impact of genomic selection in forest tree breeding





Key message

The science behind genomic selection has advanced rapidly; however, translating that knowledge into operational impact remains challenging.

PART II: Operational Pedigree to Genomics

Genomic prediction became operationally feasible.

Integrating pedigree and genomics

Cappa et al. (IUFRO Forest Tree Breeding Conference, 2014) – E. grandis

One of a major practical challenge: thousands of valuable individuals in the pedigree remained non-genotyped.

The process

1 A: expected pedigree relationships

- Based on recorded ancestry
- Captures between-family (parental average) variation

2 G: realized genomic relationships

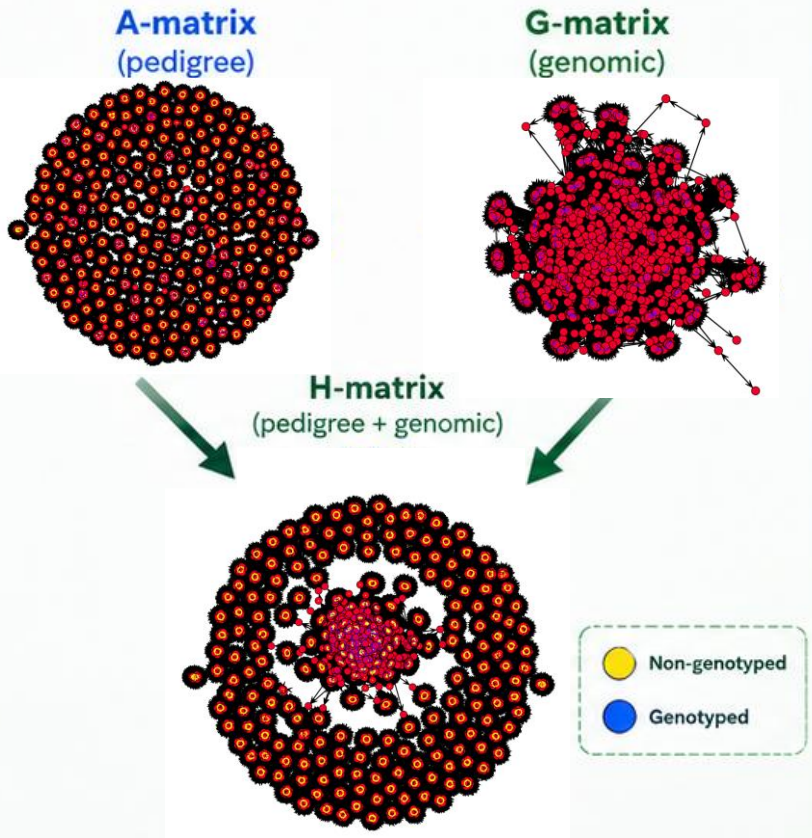
- Based on DNA markers
- Captures within-family (Mendelian sampling) variation

3 H: integrated relationship structure

- Combines A and G
- Enables joint evaluation of genotyped and non-genotyped individuals

Genotyped and non-genotyped individuals are evaluated jointly.

From A and G to *H*



Mistal et al. 2009; Legarra et al. 2009; Christensen and Lund 2010

MODEL: HBLUP (single-step GBLUP)

Joint use of phenotypic, pedigree and genomic information

$$\begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + H^{-1}\alpha \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{g} \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix}$$

Matrix of coefficients | Solutions | RHS

H matrix

$$H = \begin{bmatrix} \text{Var (Non-genotyped parents and offspring)} & \text{Cov (Non-genotyped parents and offspring, Genotyped offspring)} \\ \text{Cov (Non-genotyped parents and offspring, Genotyped offspring)} & \text{Var (Genotyped offspring)} \end{bmatrix}$$

Inverse of *H*

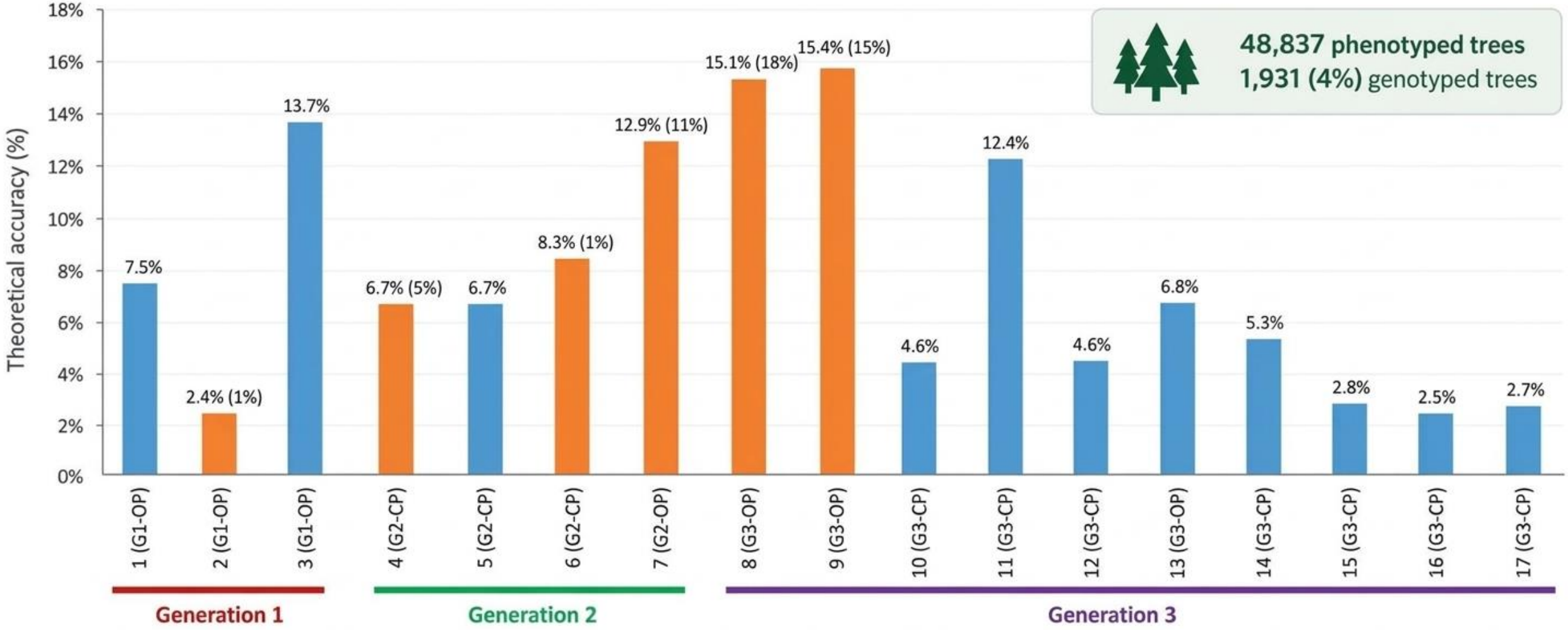
$$H^{-1} = A^{-1} + \begin{bmatrix} 0 & 0 \\ 0 & G^{-1} - A_{22}^{-1} \end{bmatrix}$$

Aguilar et al. 2010

This formulation of H^{-1} is computationally efficient.

The *H*-matrix combines both pedigree-based parental averages (between-family variation) and marker-based Mendelian sampling effects (within-family variation).

Genomic information enhances theoretical accuracy



Key message

Genomic relationships, corrected pedigrees, and ssGBLUP improve breeding value accuracy.

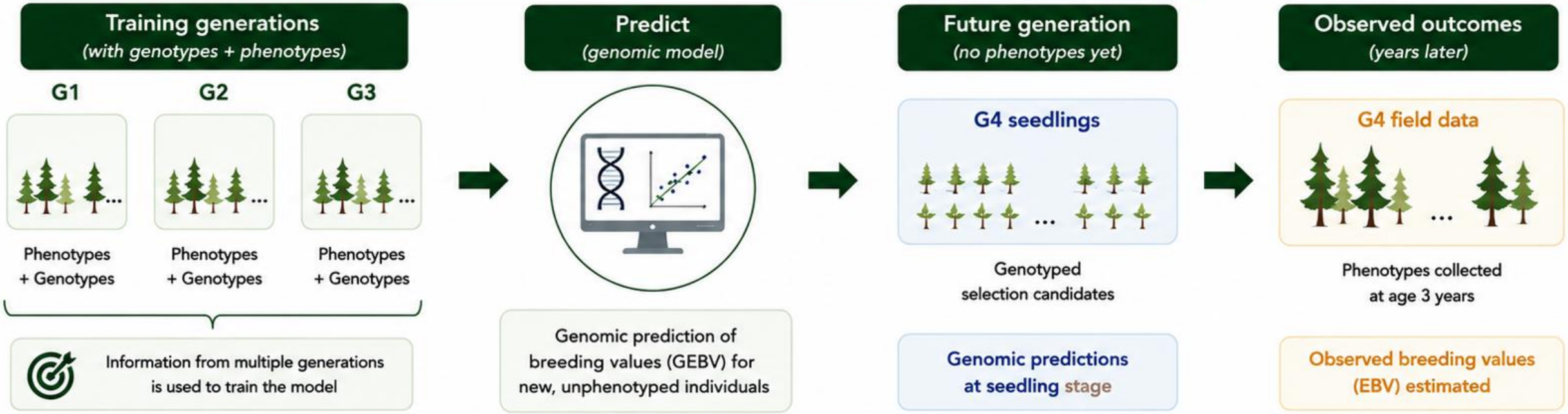
Does genomic prediction really work across generations?

Duarte et al. (Frontiers in Plant Sci, 2024) – E. grandis

Cross-validation is useful...

...but does genomic prediction work in future generations?

Accurate assessments of prediction accuracy, can only be made after the completion of the GS cycle



Realized Predictive Ability (RPA)

Correlation between genomic predictions (GEBV) in G4 and later-observed breeding values (EBV).

What makes a good training population?

Duarte et al. (Frontiers in Plant Sci, 2024) – E. grandis

Does a larger training population always produce better predictions?

Not necessarily.

Training population A
(Large but distantly related)

1,918 trees
(from three generations)

★★☆☆☆

Relatedness

Lower → RPA

VS.

29× fewer trees

Higher prediction accuracy
(for wood quality traits)

VS.

Training population B
(Closely related)

67 direct ancestors
(parents + grandparents + great-grandparents)

★★★★★

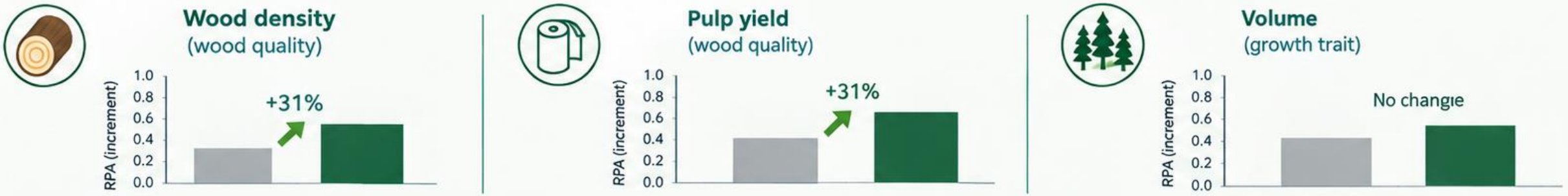
Relatedness

Higher → RPA

Why?

Close relatives share more informative **genetic** relationships with selection candidates.

Realized Predictive Ability (RPA) *across traits and training populations*



Key message

High relatedness outperforms population size in training sets.

Realized predictive ability (RPA): different traits, different challenges

Duarte et al. (Frontiers in Plant Sci, 2024) – E. grandis

RPA across traits

	Volume (VOL) Growth trait	Wood density (WD) Wood quality trait	Pulp yield (PY) Wood quality trait
RPA (individual-tree)	0.11	0.43	0.72
RPA (between-family)	0.16	0.42	0.75

GS is much more effective for wood quality traits.

RPA (individual-tree): **PY (0.72)** > **WD** (0.43) >> **VOL** (0.11)

RPA (between-family): **PY (0.75)** > **WD** (0.42) >> **VOL** (0.16)

Why is volume (VOL) so difficult to predict?

Lower heritability
Less genetic variation is captured, reducing predictive ability.

Highly polygenic
Many genes with small effects make prediction more challenging.

Stronger dependence on genome-wide LD
Prediction relies more on LD patterns, which may not transfer well.

Family selection is particularly advantageous for difficult-to-predict traits.

Two-stage genomic selection

Select families first → then individuals within top families.

Higher selection intensity and lower genotyping costs.

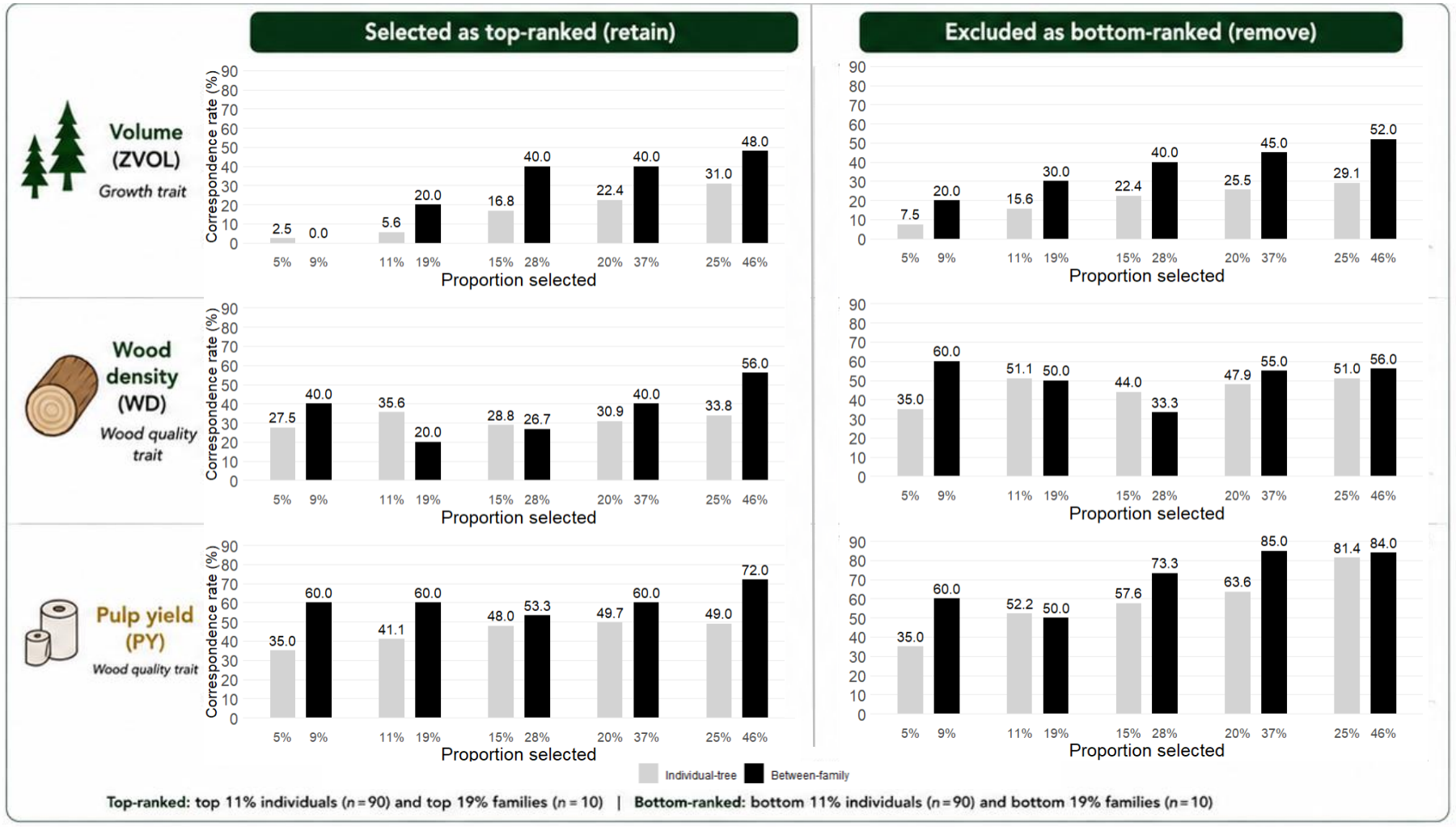
Key message

GS works, but performance depends on the trait.

Can genomic selection identify the right trees?

Duarte et al. (Frontiers in Plant Sci, 2024) – E. grandis

Correspondence between genomic (GEBV) selection and phenotypic (EBV) selection across selection intensities



Wood quality traits

GS accurately identifies superior trees and families

- ✓ Wood density (WD)
- ✓ Pulp yield (PY)

Correspondence up to **60–70%** even at high selection intensity (5–11%).

Family selection

is more reliable than individual-tree selection

- ✓ Consistently higher correspondence at all selection intensities.
- ✓ At 5% selected, ~60% of top-ranked families for WD and PY are captured.

Supports a two-stage genomic selection strategy: families first, then individuals within top families.

Early culling advantage

GS is even better at removing the worst performers.

- ✓ Higher correspondence for bottom-ranked trees and families.
- ✓ Critical for optimizing resources early in the breeding cycle.

GS is not only useful for choosing best trees.

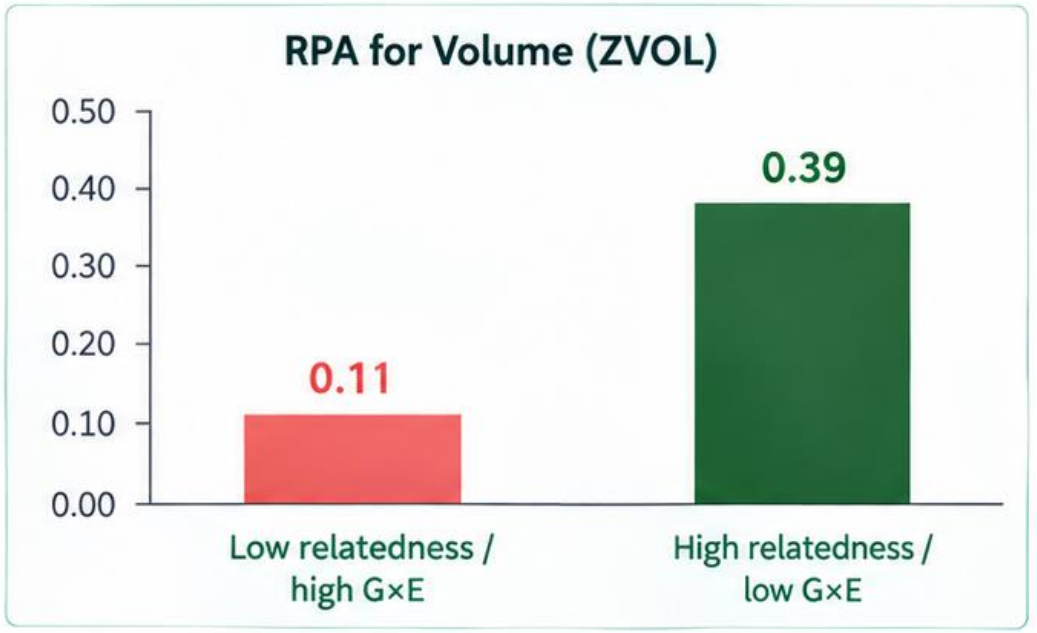
Why is prediction for volume growth so challenging?

Duarte et al. (Frontiers in Plant Sci, 2024) – E. grandis

Key training-to-testing scenarios for volume growth (Trial 8 = testing)

Scenario	Relatedness (traing → testing)	G×E	RPA
Low relatedness / high G×E	Low	High	0.11
High relatedness / low G×E	High	Low	0.39

~3× higher RPA



1 Relatedness matters
Closer training–testing relationships improve RPA.

2 Environmental similarity matters
Low G×E improves prediction transfer across sites.

3. Model and training set matter
ssGBLUP + direct ancestors + diverse training sets improve RPA.

Genomic selection is mature as a statistical methodology.

It's operational success depends on understanding the breeding population, the trait, the data and the biological context.

PART III: Multi-trait Biological Integration

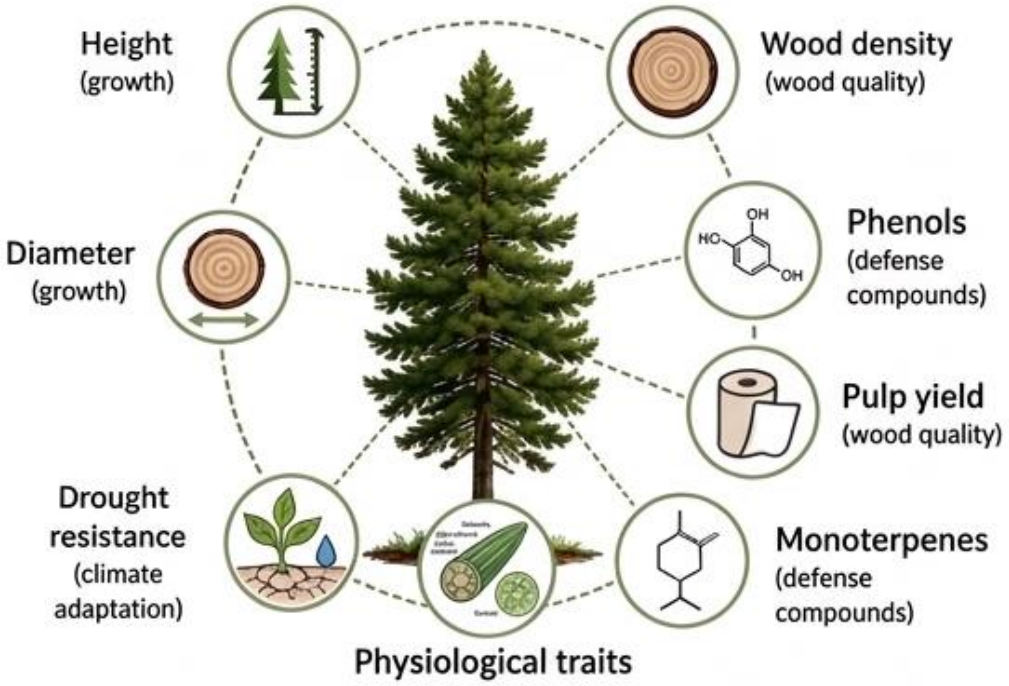
The fourth limitation: traits do no exist independently

Cappa et al. (BMC Gen., 2022) – Lodgepole pine

Traits are not independent: they share biological functions and genetic information.

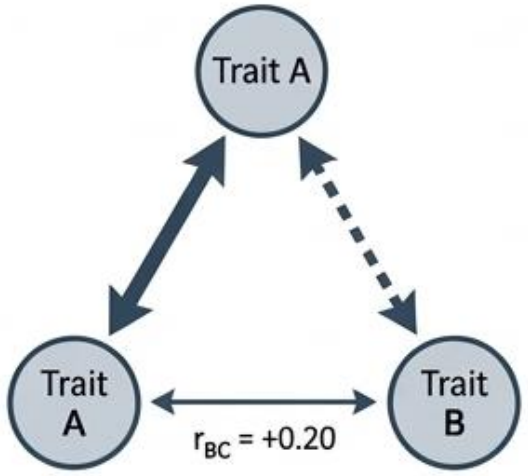
1. BIOLOGICAL SYSTEM

Traits are connected through biology and physiology.



2. GENETIC RELATIONSHIPS

Traits are genetically correlated and share information.



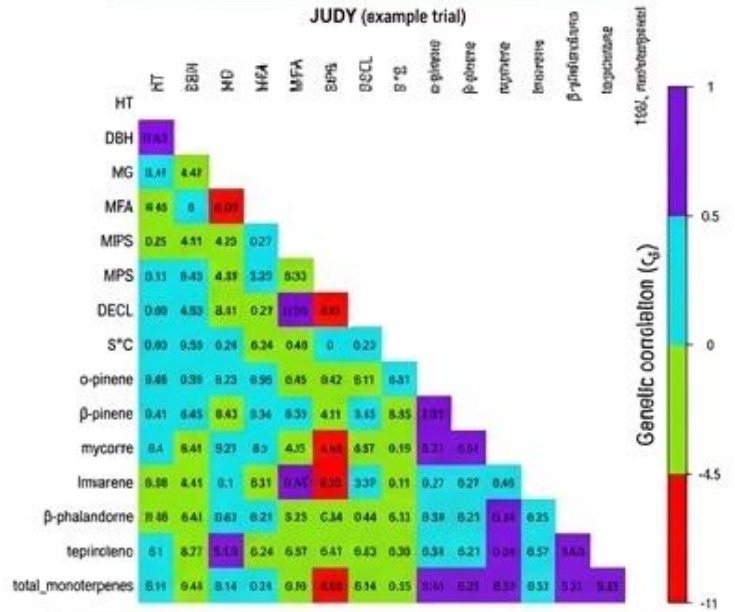
Genetic correlations

3. INFORMATION SHARING AND PREDICTION

Multi-trait model

Borrow information across genetically correlated traits.

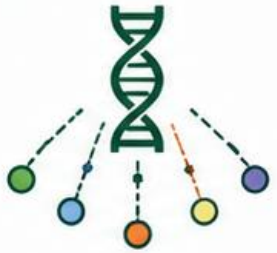
Traits are genetically correlated



Genetic correlations create opportunities to borrow information across traits.

Multi-trait models: impact on GWAS power and accuracy

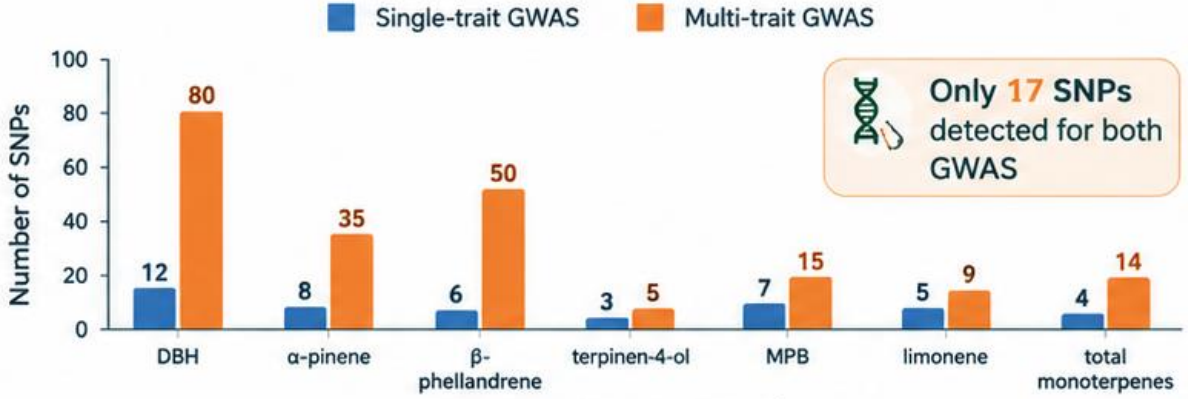
Cappa et al. (BMC Gen., 2022) – Lodgepole pine — *Jurcic et al. (TGG, 2023) – E. grandis*



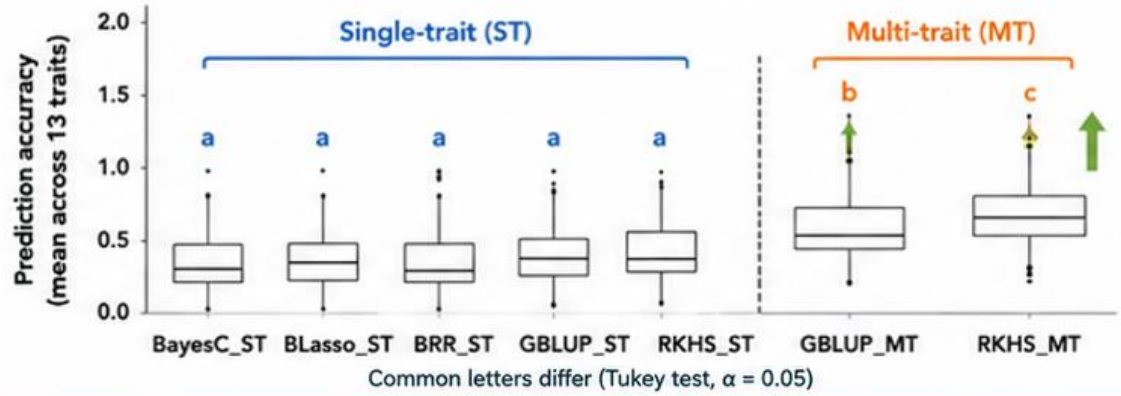
Pleiotropy

Multi-trait model

More QTL detected: increased GWAS power



Higher prediction accuracy



Predicting traits that are difficult or expensive to measure

Challenge
Wood-quality traits are expensive and often sparsely recorded.

Multi-trait strategy
Borrow information from easily measured growth traits.

Result
Higher predictive accuracy for difficult-to-**measure traits.**

Up to +17.7%
in predictive accuracy

*Multi-trait models can **extend genomic prediction** to traits that are costly or **difficult** to measure.*

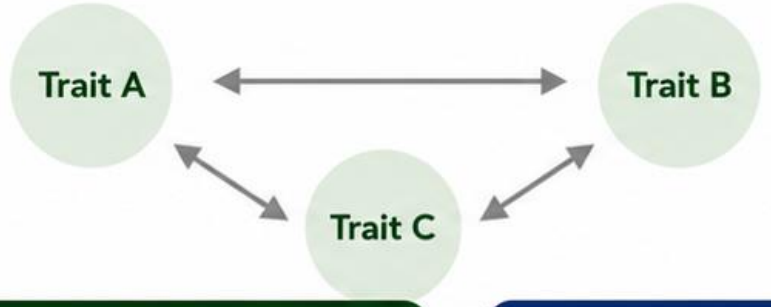
Key message

Borrowing information across traits enhances biological discovery, improves prediction, and enables prediction of difficult or expensive recorded traits.

PART III: From Correlation to Causation

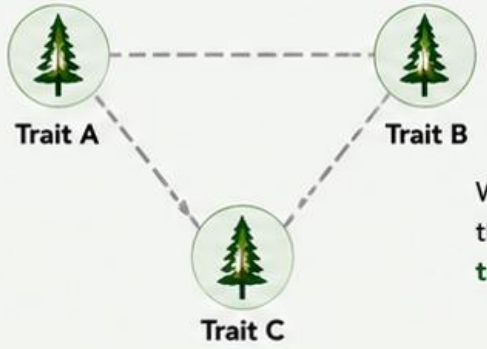
The fifth limitation: correlation is not causation

Traits can be correlated… but correlation does not tell us *why*.



? What generates these correlations?

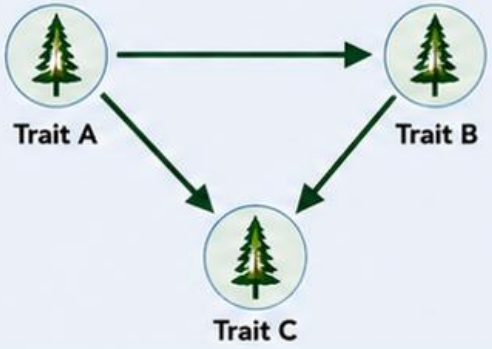
CORRELATION (what we usually see)



We only observe that traits **move together.**

VS.

CAUSATION (what we want to uncover)



One trait influences another through **direct or indirect effects.**

Why the same correlation can arise



Shared genetics
(pleiotropy / linkage)
Traits are correlated because of their shared genetic basis.



Biological regulation
Traits are correlated because one affects another through biological pathways.

Correlation ≠ Causation
They do not reveal the underlying mechanisms that generate those associations.



We need models that can **discover and quantify causal relationships among traits.**



Bayesian Networks & Structural Equation Models

Multi-trait models tell us that traits are related. → *Causal models try to explain why they are related.*

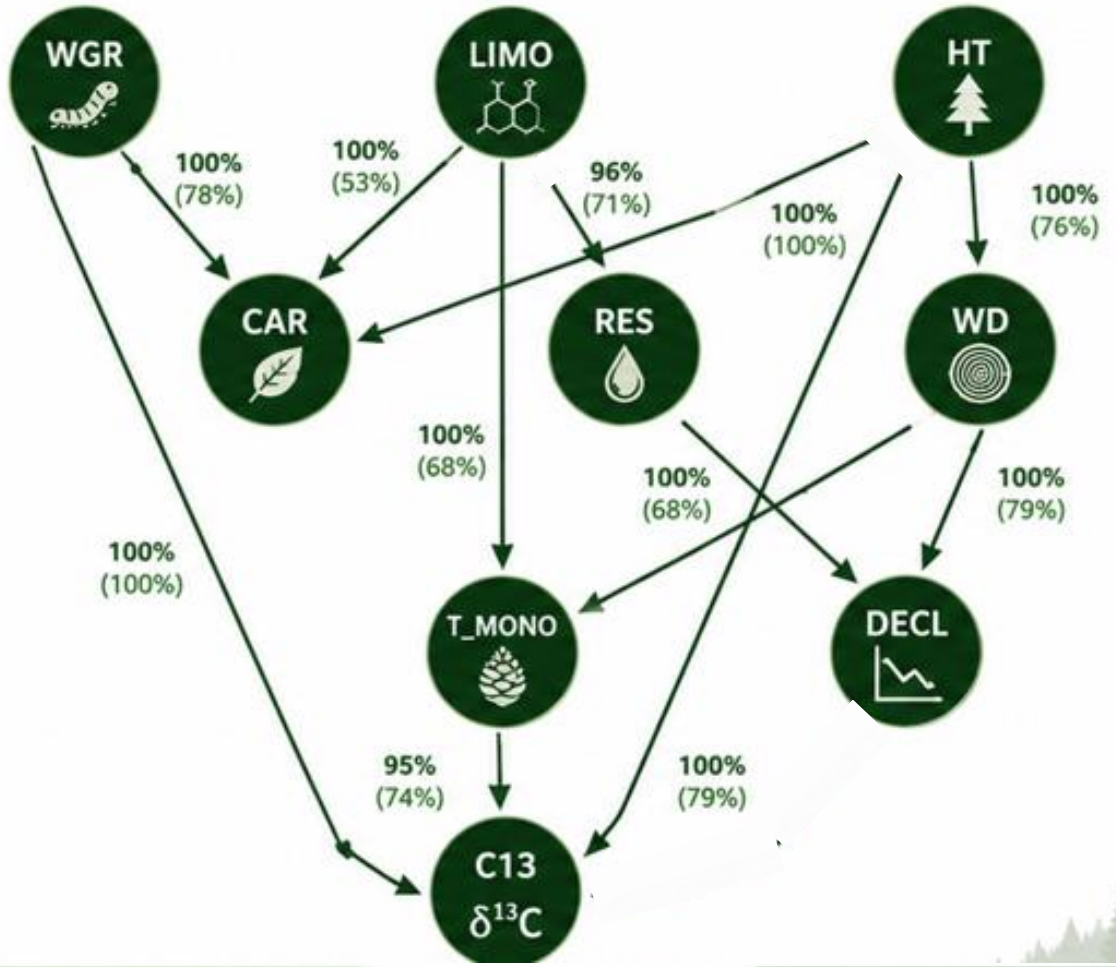
Bayesian network

Cappa et al. (G3, 2025) – Lodgepole pine

Phenotypes
e.g., growth, wood quality, defense, climate adaptation

Bayesian Network learning
Data-driven discovery of causal relationships

Directed network
Causal structure among traits (example from lodgepole pine)



- Each arrow represents a direct causal relationship
- xx% how consistently the link appears
- (xx%) how certain we are about the causal direction

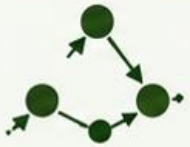
- infer direction of information flow
- distinguish direct and indirect relationships

Example: Lodgepole pine
392 trees, 25,099 SNPs
9 traits

Structural Equation Models improve prediction

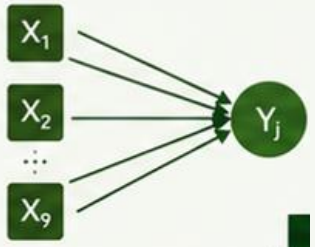
Cappa et al. (G3, 2025) – Lodgepole pine

Bayesian Network



Learns the causal structure among traits

Structural Equation Model (SEM)



Genetic effects reflect the direct impacts acting specifically on each trait

$$\boldsymbol{y}_i = \boldsymbol{\Lambda}\boldsymbol{y}_i + \boldsymbol{X}_i'\boldsymbol{\beta}^* + \boldsymbol{a}_i^* + \boldsymbol{e}_i^*$$

$\boldsymbol{\Lambda}$ is a 9 × 9 lower-triangular matrix of structural equation coefficients (λ).

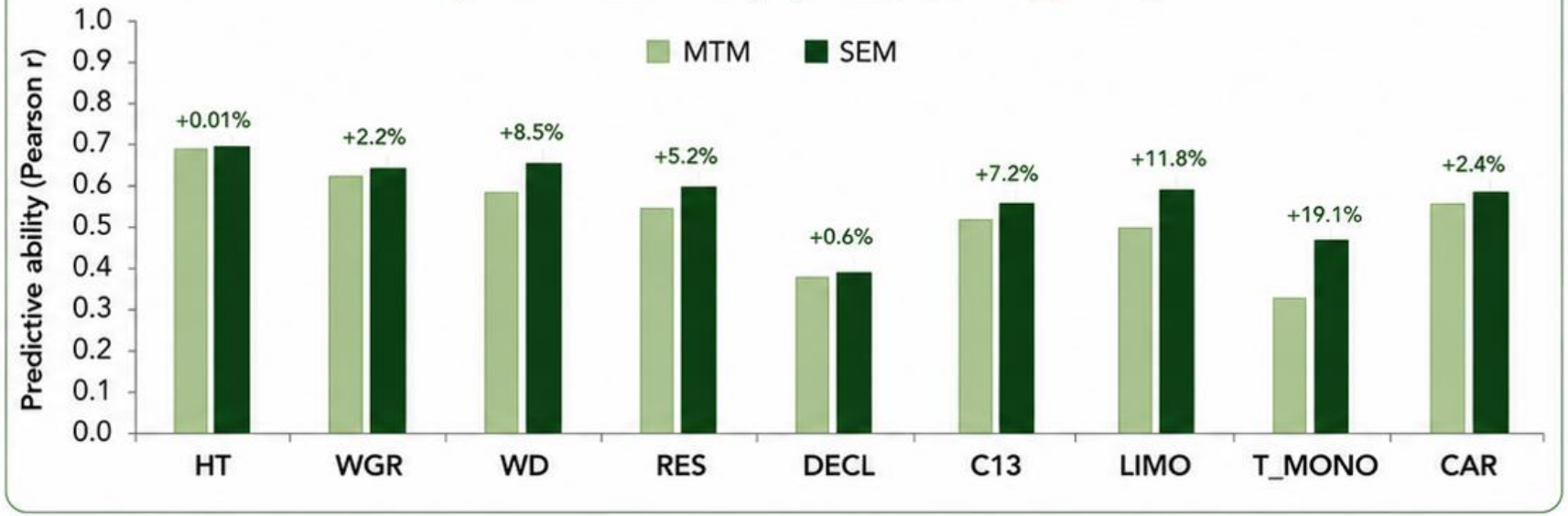
SEM outperforms traditional multi-trait (MTM) models

↓ Residual variance
Average reduction (MTM → SEM)
8 – 21%

↑ Heritability
Average increase (MTM → SEM)
9 – 39%

↑ Predictive ability
Average gain in accuracy (MTM → SEM)
6%

Example: Predictive ability by trait (GBLUP approach)



Main message | Accounting for causal relationships produces biologically more realistic breeding value predictions.

Example: Lodgepole pine
392 trees, 25,099 SNPs
9 traits (growth, defense and climate adaption)

PART III: Machine Learning for Genomic Prediction

The seventh limitation: beyond linear models

Cappa et al. (TGG, 2026) – White spruce



Linear ssGBLUP

↓

assumes

Additive relationships

↓

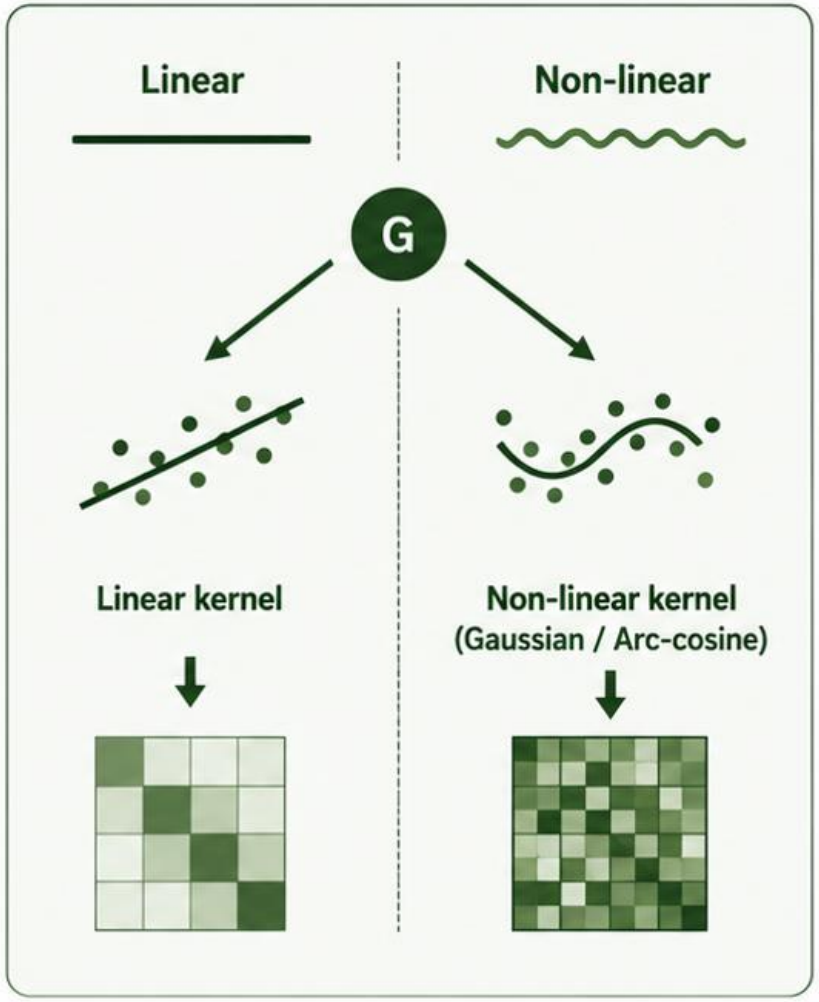
Complex forest traits are rarely driven by purely additive effects alone.

↓

Machine Learning kernels

↓

capture complex genetic patterns



Can Machine Learning kernels improve genomic prediction by capturing non-linear genetic architectures?

Machine learning kernels improve genomic prediction

Cappa et al. (TGG, 2026) – White spruce

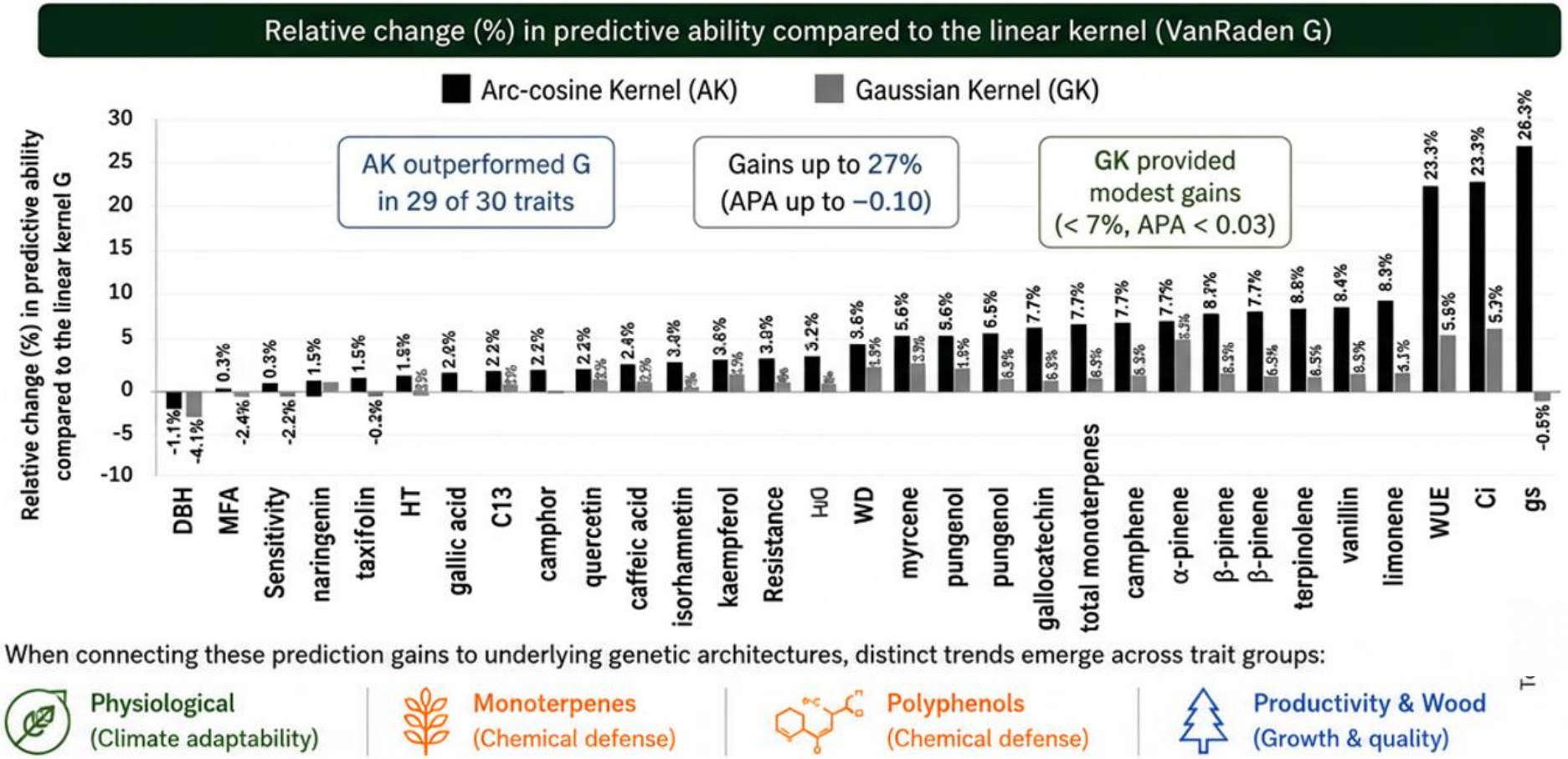
White spruce (30 traits) – Single-step GBLUP models

KERNELS EVALUATED

VanRaden (G)
Linear kernel
Captures additive genetic relationships.

Gaussian Kernel (GK)
Non-linear kernel
Models similarity that decays with genetic distance.

Arc-cosine Kernel (AK)
Deep non-linear kernel
Captures complex, higher-order non-linear interactions.



When connecting these prediction gains to underlying genetic architectures, distinct trends emerge across trait groups:

- Physiological (Climate adaptability)
- Monoterpenes (Chemical defense)
- Polyphenols (Chemical defense)
- Productivity & Wood (Growth & quality)

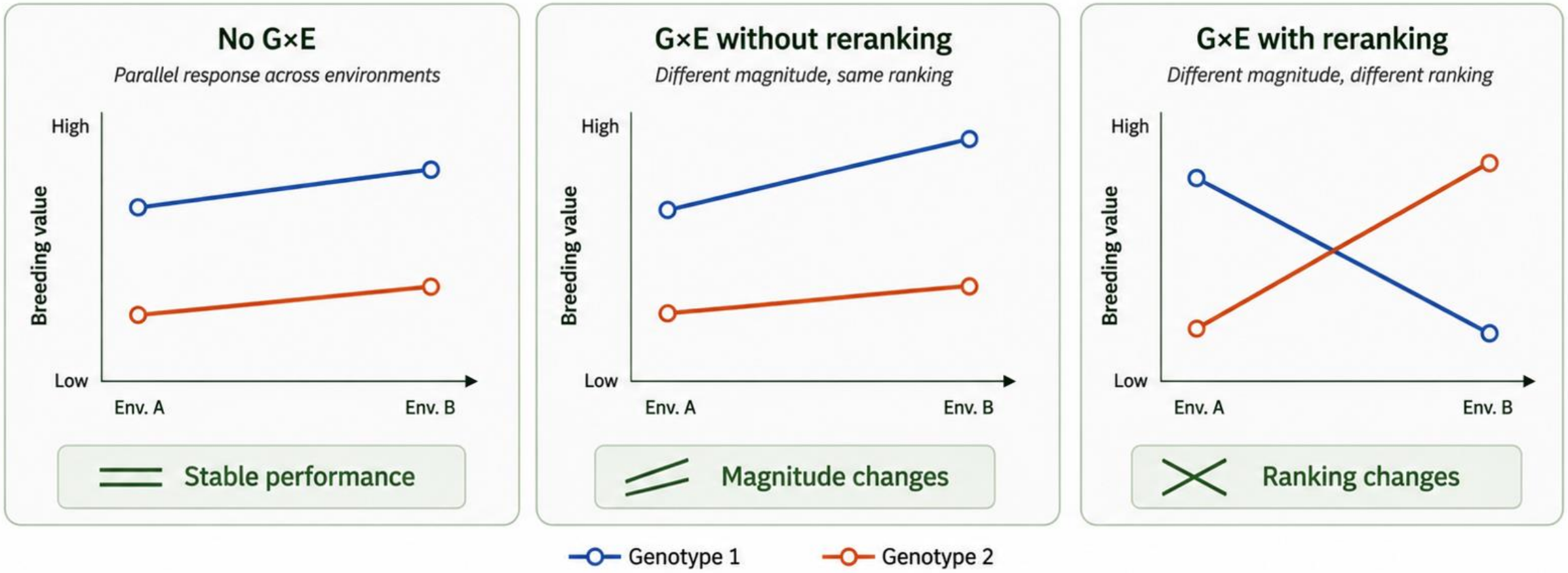
Key message

Prediction gains are context-dependent and depend on both the genetic architecture of the trait and the choice of kernel.

PART IV: Prediction Across Environments

Prediction across environments

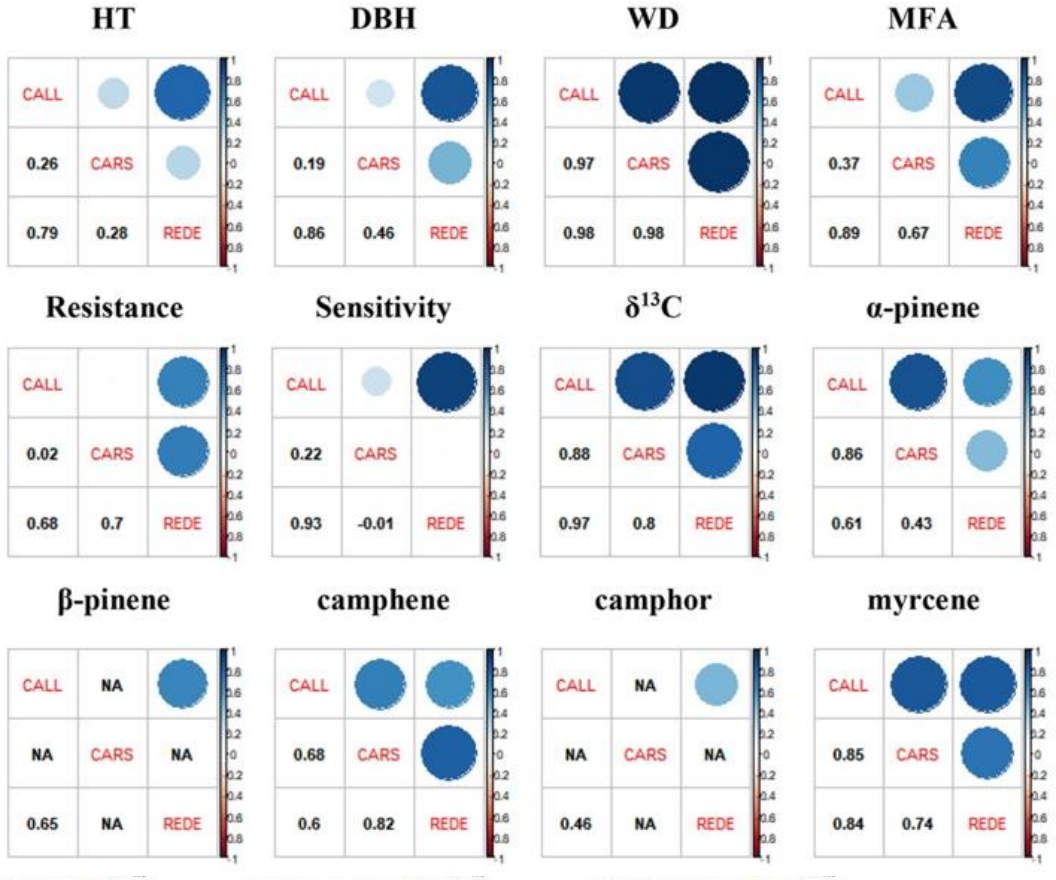
Genetic merit is not always the same across environments



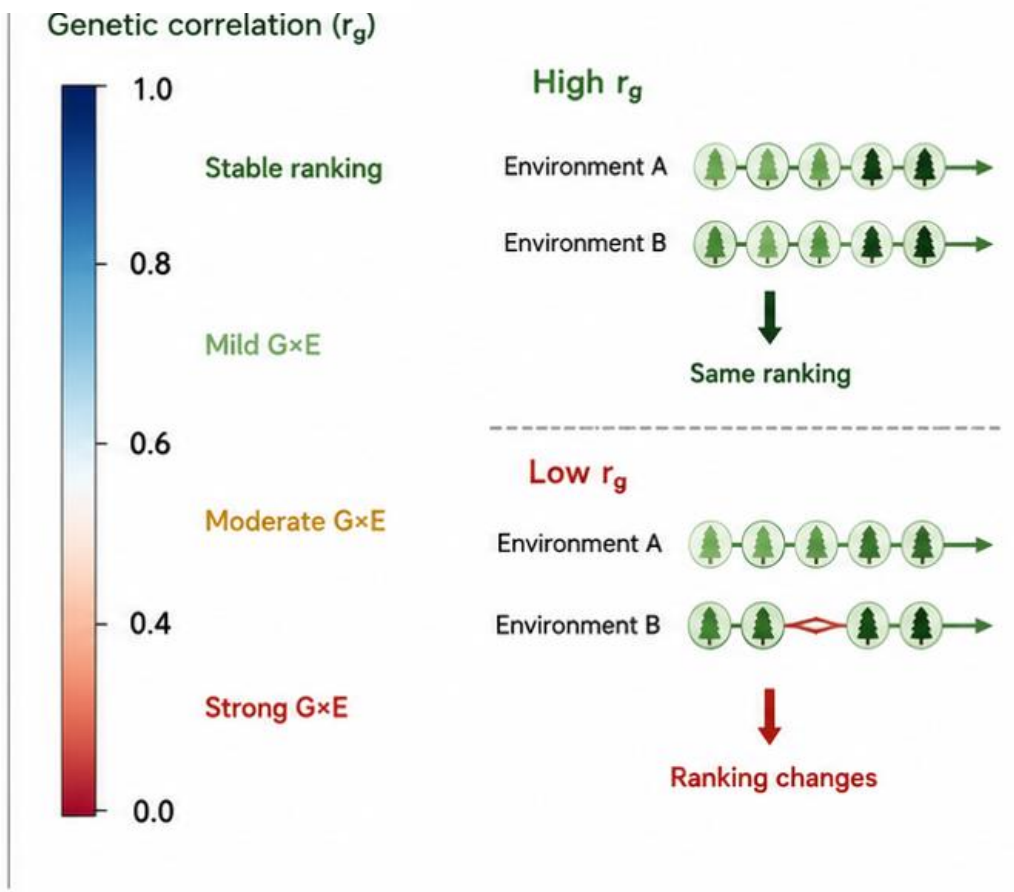
Measuring G×E using genetic correlations

Cappa et al. (PLOS One, 2022) – White spruce

Estimated genetic correlations between sites for 12 (15) traits (White spruce population).



Upper diagonal shown. Circle color and size reflect the strength of genetic correlation (r_g).
NA = not estimated at the CARS site due to insufficient phenotypic data.



Key message | **Genetic correlations** provide a direct measure of G×E.

Beyond discrete environments

Traditional multi-environment models



Discrete covariance matrix (example)

	A	B	C	D
A	1.00	–	–	–
B	0.82	1.00	–	–
C	0.55	0.63	1.00	–
D	0.31	0.44	0.68	1.00

Finite number of environments

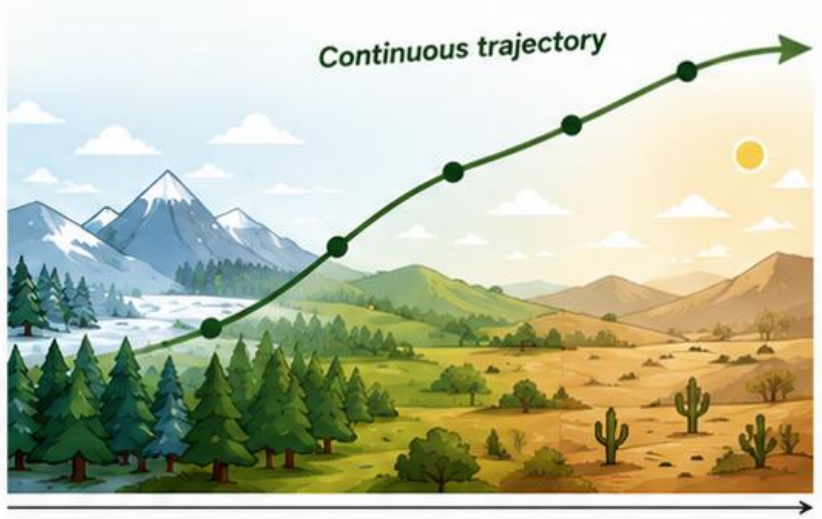
We lose valuable information, and we can only estimate genetic relationships between tested locations.

From discrete environments

to continuous environmental variation

Continuous environmental gradients

- Temperature
- Precipitation
- Water deficit
- Altitude
- Soil properties (e.g., texture, fertility, pH)



Random regression & covariance functions allow breeding values and covariances to change continuously across environmental gradients.

Key message

Genetic prediction should follow environmental gradients, not site categories.

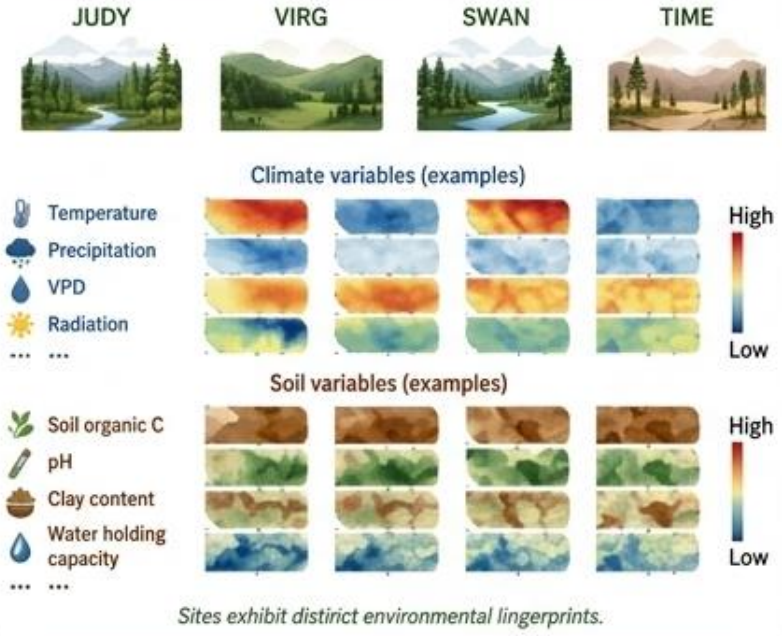


Explaining G×E using environmental covariates

Cappa et al. (In preparation, 2026) – Lodgepole pine

Using climate and soil covariates to model environmental similarity and explain site variance.

1. Environments, different locations, different conditions



Sites exhibit distirict environmental lingerprints.

2. Environmental covariates

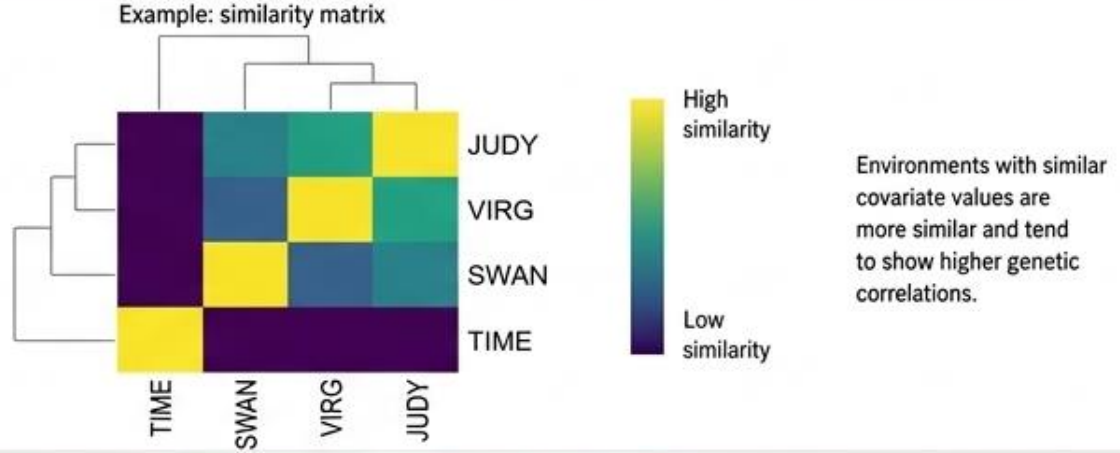
Quantify environments using measurable descriptors

Climate covariates
- Temperature (mean, min, max)
- Precipitation
- Vapor pressure esfect (VPD)
- Radiation
- Growing degree deys
- ...

Soil covariates
- Soil organic carbon
- pH
- Soil texture (tand, clay)
- Water holding capacity
- Soil fertility
- ...

From descriptors to environmental similarity

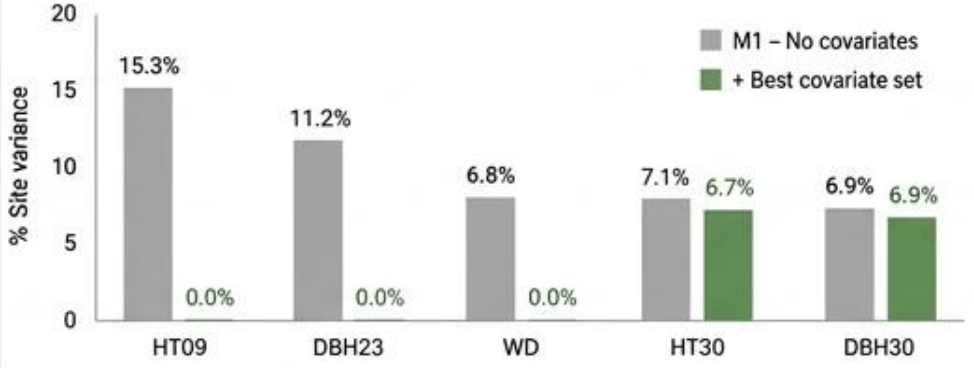
3. Environmental similarity among sites



4. Environmental covariates explain site effects

Adding covariates reduces the (co)variance attributed to sites

Examples: % of phenotypic variance due to site



Examples

HT09
Site effect reduced from **15.32%** to **0.00%** with soil covariates.

WD
Climate covariates reduced site effect to **0.00%**.

HT30 / D8H30
Site variance remained virtually unchanged.

Key message

Environmental covariates capture environmental differences among sites and reducing the site effect.

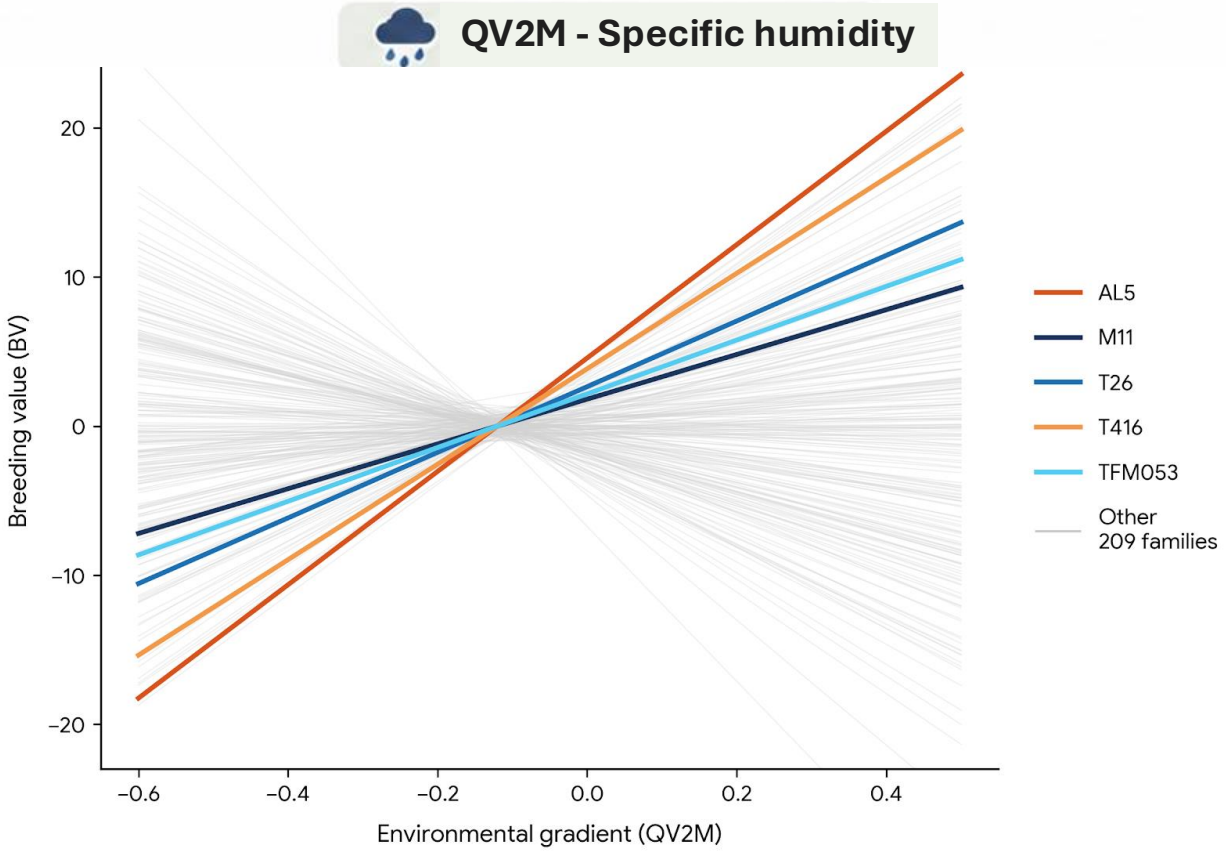
Different traits are influenced by different environmental drivers.

Predicting breeding values across environmental gradients

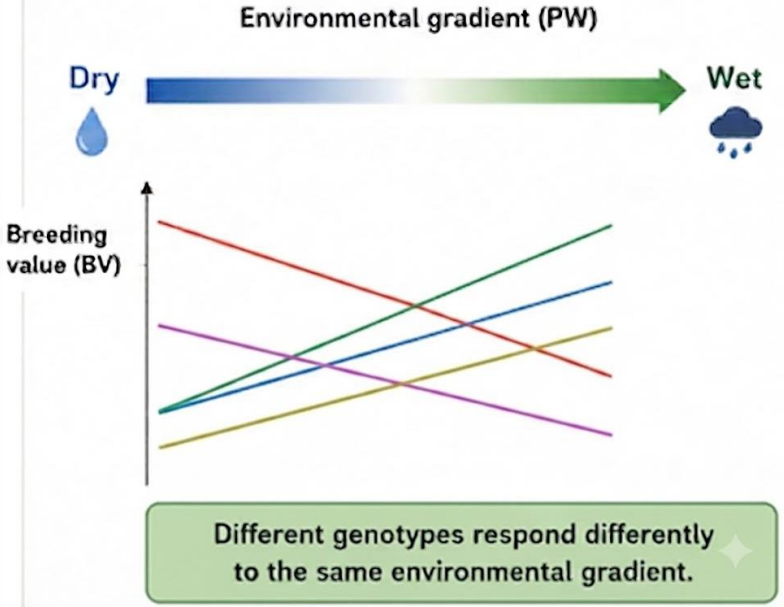
Belaber et al. (In preparation, 2026) – Pinus taeda

Reaction norm models describe how breeding values change along continuous environment gradients

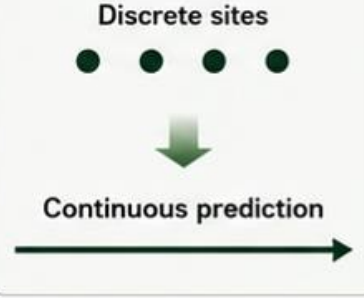
Reaction norm for 214 families along the specific humidity at 2 m. (QV2M) gradient



What do reaction norms represent?



From discrete sites to continuous prediction



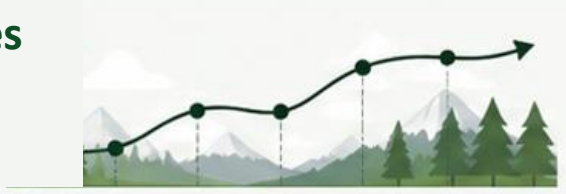
Conceptually:

$$BV_i(E) = a_i + b_i E$$

Breeding values are modeled as functions of continuous environmental variation.

Key message

Reaction norms enable continuous prediction of breeding values across environmental gradients, including both observed and unobserved environments.



Beyond climate and soil: adding targeted monoterpenes

Cappa et al. (In preparation, 2026) – Lodgepole pine

Targeted metabolomics

Biomarkers of adaptation and G×E

- Captures genotype × environment responses
- Adds biological information beyond climate and soil
- Targets specific metabolites linked to plant response



Climate + Soil

+

Targeted metabolomics

↓

Better prediction in unobserved environments

Impact on predictive ability in unobserved environments

Relative change (%) in predictive ability vs. baseline model (*M1*)

Covariate set	Linear kernel				Gaussian kernel (AK, *h* = 1)				Arc-cosine kernel (AK, 1 layer)			
Model name	JUDY	VIRG	SWAN	TIME	JUDY	VIRG	SWAN	TIME	JUDY	VIRG	SWAN	TIME
Climate	0.14	0.17	5.81	2.32	0.18	0.10	5.94	2.47	3.38	0.21	5.77	2.32
Soil	0.07	0.07	5.60	2.44	0.14	0.00	2.74	2.44	0.07	0.17	5.56	2.44
Targeted monoterpenes	0.14	0.17	5.56	2.32	0.04	0.10	5.52	2.47	0.21	0.10	**7.85**	2.44

Largest improvement: +7.85%

SWAN × targeted monoterpenes × AK kernel

Targeted metabolomics adds biological information that climate and soil alone cannot capture.

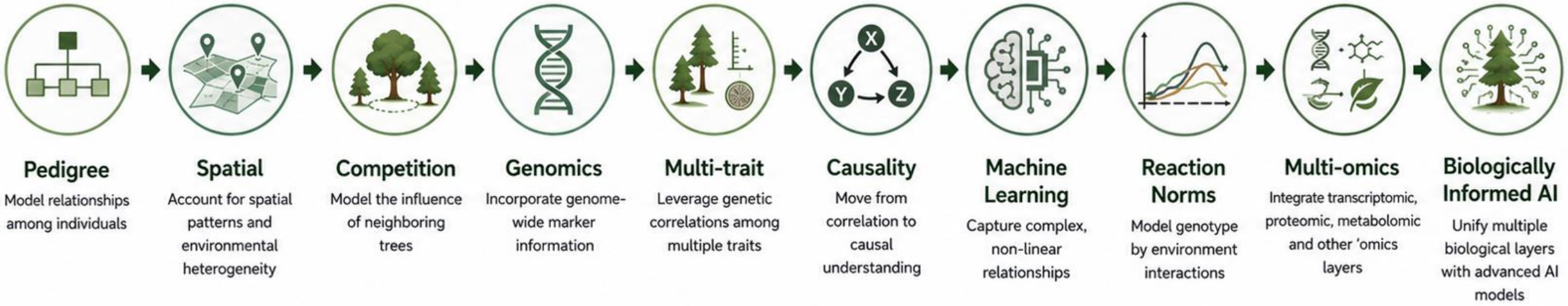
Climate + Soil + Biology → **Better prediction**

PART IV: Toward Biologically Informed Prediction Models

Looking ahead in forest tree breeding

The evolution: 20 years of increasing biological realism

Over the last twenty years, we have progressively incorporated new sources of biological information into prediction models.





All new methods share the same goal:
building models that better capture real biology.

From statistical prediction to biologically informed prediction

What's next?

Not just: Adding more data or complex statistical models
The real goal: Integrating biological data into a unified predictive framework

Statistical genomics-centered prediction approach

Pedigree	Genomics	Phenotypes
Relationships among individuals	Genome-wide markers (SNPs)	Trait measurements in testing environments

Limited biological information

We move toward...

Multi-omics integrated biologically informed prediction approach

Genomics	Enviromics	Transcriptomics	Metabolomics	Proteomics	Artificial Intelligence
Genetic variants and structure	Climate, soil and other ecological factors	Gene expression and regulatory activity	Metabolites and biochemical pathways	Proteins and cellular functions	Algorithms to learn complex patterns across layers

Multiple biological layers interact to shape complex traits

Better biological representation → Better prediction → Better breeding decisions

The future of forest tree breeding is not about building more complex statistical models.

It is about building biologically informed prediction models.

International Research Collaborations



Always open to new collaborations.

Thank you for your time!

Check out all full papers and the presentation here!

My contact

Dr. Eduardo Pablo Cappa

Bosques Cultivados

https://www.linkedin.com/in/eduardo-pablo-cappa-a29b7a2b9/

